Bridging Bangla Dialects and Standard Bangla: A Dialect-Aware Bangla Question Answering with Routed Domain Experts and Retrieval-Augmented Generation for Healthcare and Education

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A thesis submitted to the Department of Computer Science and Engineering
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B.Sc. in Computer Science and Engineering

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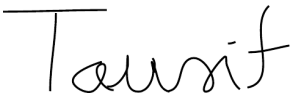
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2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Abstract

There are many Bangla speakers who communicate using regional variations such as Sylhet and Chittagonian. While existing Bangla question answering systems and language models are developed and evaluated on Standard Bangla, this creates a particular gap in the Healthcare sector and education sector in the domain where reliable information is important. This study proposes a dialect aware multi domain Bangla QA framework based on router driven mixture of expert architecture. These frameworks collect questions in Bangla dialect or standard Bangla and responses according to that input. A light weight routing model first recognises standard Bangla and linguistic variety of the input and its domain. These inputs and questions are then normalized into standard Bangla if it is recognized by the model as a dialect before being routed to the proper domain so that each domain is supported by a dedicated retrieval augmented generation (RAG) pipeline so that it matches the response generation in relevant external evidence. There will be multiple predetermined language models where the framework compares several open source Bangla adapted TigerLLM and other suitable architectures. These models are adapted using efficient fine tuning methods such as LoRA and QLoRA. The framework is determined using Bangla question answering and domain specific resources including NCTB-QA, BanglaRQA and BanglaCHQ. The evaluation covers dialect identification and normalization, retrieval effectiveness, end-to-end QA performance, clarification and abstention behavior. Then the results are separately reported across that specific dialect language or standard Bangla with component wise studies and comparisons across candidate base models. This study aims to reduce the performance disparity between standard Bangla and dialect users by developing a modular retrieval, grounded and domain hour QA framework that provides standard Bangla responses while being computationally practical for low resource Bangla NLP.

Keywords: Bangla NLP, Bengali medical chatbot, Bengali educational chatbot, Chittagonian, Sylheti, dialect normalization, dialect routing, question answering, retrieval-augmented generation, medical safety, Gemma 4, TigerLLM, LoRA, QLoRA, mixture of adapters, NCTB-QA, BanglaRQA, BanglaCHQ, low-resource NLP

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Nomenclature

The following abbreviations are used in this thesis.

ASR	Automatic Speech Recognition
BLEU	Bilingual Evaluation Understudy
BM25	Best Matching 25 (sparse retrieval function)
CER / WER	Character / Word Error Rate
chrF / chrF++	Character n-gram F-score (++: with word n-grams)
CI	Confidence Interval
Ctg / Syl	Chittagonian / Sylheti
DEV / TEST	Validation split / held-out test split
ECE	Expected Calibration Error
EM	Exact Match
KB	Knowledge Base
LLM	Large Language Model
LoRA	Low-Rank Adaptation
MCQ	Multiple-Choice Question
MoE	Mixture of Experts
MRR	Mean Reciprocal Rank
NCTB	National Curriculum and Textbook Board
nDCG	Normalized Discounted Cumulative Gain
NF4	4-bit NormalFloat quantization
QA	Question Answering
QLoRA	Quantized Low-Rank Adaptation
RAG	Retrieval-Augmented Generation
RRF	Reciprocal Rank Fusion
SVM	Support Vector Machine
TF-IDF	Term Frequency–Inverse Document Frequency

Chapter 1

Introduction

Bangla is spoken by more than 270 million people [103], but most language technology for Bangla understands only one form of it: Standard Bangla. As with many languages spoken by many people, naturally, there are many variations to how Bangla is spoken. These variations are dialects and this creates a communication barrier between people from different regions. At home and online, many people write the way they speak, in a regional dialect such as Chittagonian or Sylheti, or type Bangla with Latin letters (romanized Bangla). This chapter explains why that matters for question answering (QA) in healthcare and education, states the problem, the research questions and the objectives, summarises the method and the contributions, and sets out the scope and the main challenges of the work.

1.1 Background

Ranking among the world's most prominent languages by total speakers, Bangla has a significant global presence; across Bangladesh, India and the diaspora, millions of people use it as their first or second language [75], [103]. Bangla has a strong regional dialect continuum: Bangladesh alone has about 55 regional varieties spread over its 64 districts [51], [74]. These varieties differ from Standard Bangla (cholito bhasha), the form used in textbooks, news and administration, in pronunciation, vocabulary, verb morphology and word order. Many speakers learn a regional variety at home and Standard Bangla only at school. Chittagonian and Sylheti are two of the most widely spoken and most divergent varieties: in dialect translation and dialect identification experiments they are consistently among the hardest to handle [103], [51], [70], [86]. In our own data, a Chittagonian sentence and its Standard translation share only 30% of their character trigrams, and a Sylheti sentence 35% (Chapter 4).

Resources for natural language processing (NLP) are unevenly distributed across the world's languages, and low-resource languages receive only a small share of data, models and benchmarks [29]. Bangla has gained important resources in recent years: pretrained encoders and sequence-to-sequence models such as BanglaBERT and BanglaT5 [40], [49], Bangla large language models (LLMs) such as TigerLLM and TituLLMs [90], [88], and QA benchmarks such as BanglaRQA, NCTB-QA, BEnQA and SOMAJGYAAN [41], [99], [75], [92]. All of these, and the textbooks,

news articles, encyclopedia passages and exam papers they are built from, are in Standard Bangla, and models are evaluated only on Standard Bangla [103]. Figure 1.1 summarises this situation.

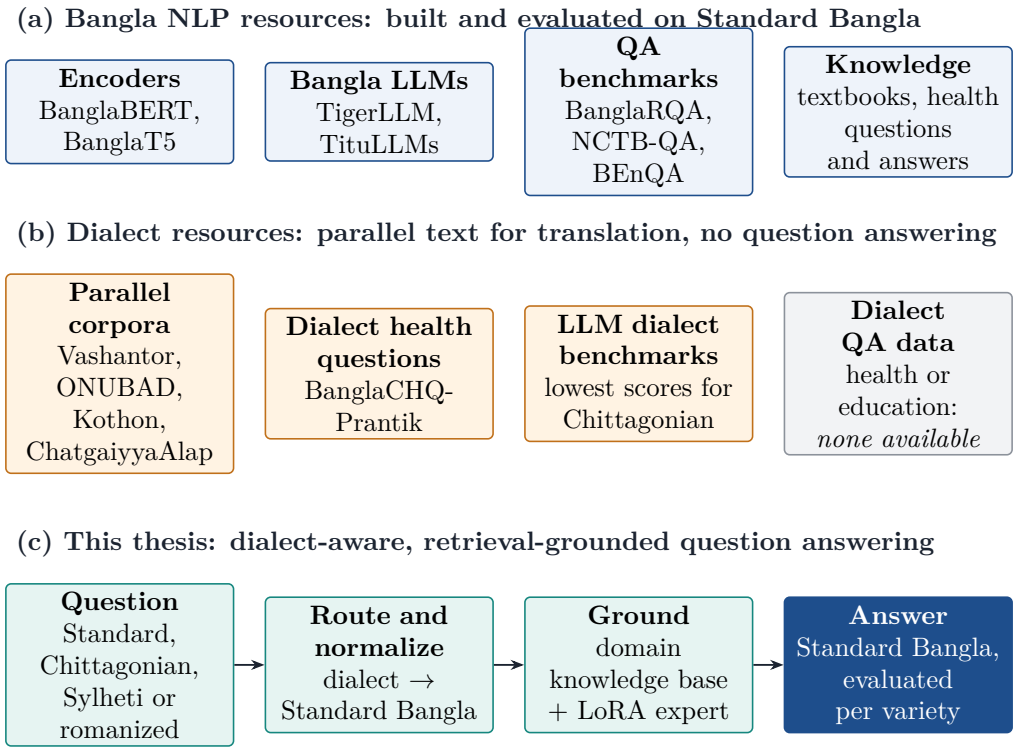


Figure 1.1: (a) Bangla language technology is built on Standard Bangla, while (b) dialect resources support translation but not question answering. (c) This thesis connects the two: dialect questions are normalized into Standard Bangla so that existing Standard-Bangla QA data and knowledge sources can be reused.

Large language models have made open-ended question answering widely accessible, but they can produce fluent answers that no evidence supports, a behaviour known as hallucination [53]. Retrieval-augmented generation (RAG) reduces this risk by retrieving relevant documents at answer time and conditioning the generator on them [33]. Parameter-efficient fine-tuning with low-rank adaptation (LoRA) and its quantized variant QLoRA makes it possible to adapt billion-parameter models on a single GPU [43], [50]. The idea of dividing a task among specialised sub-models chosen by a gating or routing function goes back to the mixture-of-experts (MoE) literature [3], [18]. Together, these techniques make a modular, grounded and affordable QA system realistic for a low-resource language.

The dialect problem matters most where reliable information is scarce. In **healthcare**, Bangladesh has about 9.9 doctors, nurses and midwives per 10,000 people, and only about 11% of doctors and nurses work in rural areas, where about two thirds of the population live [45]. Digital health tools mostly serve European users and languages, and a Bangladeshi triage application reported that a general speech-to-text service struggled with dialect speech [87]. Real consumer health questions in Bangla are informal, loosely described and code-mixed with English medical terms [56], Bangla telemedicine consultations are full of vague descriptions, delayed disclosure and regional expressions, and prioritising patients from such real consultations

is hard even for strong models (best triage macro-F1 0.463) [104]. In **education**, university students in Bangladesh already use LLMs for learning [84], and national-curriculum (NCTB) textbooks have become the basis of new Bangla QA resources and models [90], [99]. Yet LLMs answer the same exam questions less accurately in Bangla than in English [75], earlier Bangla textbook QA systems were extractive models that only worked when the correct passage was supplied [71], and no educational QA resource contains questions asked in a regional dialect.

1.2 Rationale of the Study

The dialect gap is large and prompting alone does not close it. A benchmark of 19 LLMs on questions in nine Bengali dialects found far lower scores for Chittagonian (5.44 out of 10) than for Tangail (7.68) [106]. When LLMs are prompted to convert dialect text into Standard Bangla, some score below a baseline that simply copies the input unchanged [103]. In the reverse direction (Standard Bangla to dialect), the best prompted system on dialect health questions reached a BLEU of only 23.67 for Sylheti, and Chittagonian scores were generally lower (14.78 in the same setting) [86]; without retrieved examples, LLMs failed almost completely at standard-to-dialect translation (BLEU 5–12) [91]. In our own paired evaluation, the same questions asked in dialect scored 33% (Chittagonian) and 43% (Sylheti) lower than in Standard Bangla when the system did not normalize them (Chapter 5). If a system does not understand the question, the quality of its knowledge base and generator hardly matters.

Small amounts of targeted supervision help. LoRA fine-tuning of open LLMs on only 160 examples per dialect raised Mistral-7B from 16.2 to 73.6 BLEU on dialect-to-English translation, above every prompted closed-source model [103]. A lightly adapted dialect component is therefore a realistic way to close much of the gap at low cost.

Normalizing into Standard Bangla reuses existing resources. Almost all Bangla QA data, textbooks and health information exist only in Standard Bangla, and building separate datasets and knowledge bases for every dialect is not realistic. Rewriting a dialect question into Standard Bangla first lets the system reuse all of them. There is supporting evidence for this "translate into the better-resourced variety" strategy: appending an English translation of the question improved LLM accuracy on BEnQA across all subjects [75], and dialect-to-Standard translation is the central step of Bangla speech-standardization pipelines [74].

Different domains need different evidence and safety behaviour. Health answers must be grounded in reviewed information, and the system must recognise emergencies and know when to refer the user to a doctor. Education answers should follow the curriculum, and the system must recognise questions that its sources cannot answer; NCTB-QA makes 42.75% of its questions unanswerable for this reason [99]. Routing questions to domain-specific pipelines keeps the prompts, knowledge bases and safety rules of each domain separate and easy to inspect, and small routers that choose a retrieval strategy per question are an established idea [67]. Showing users the passages behind each answer is a simple way to make the system's reasoning inspectable.

The system must be affordable. The institutions that would benefit most, such as rural clinics, schools and small organisations, cannot run very large models or pay per-question API fees. A design with CPU routers, one shared open 4-billion-parameter model and a few small LoRA adapters can be trained and served on one rented GPU [50]. This approach keeps the system easy enough to build for a student project, yet lightweight enough for constrained budgets.

1.3 Problem Statement

Bangla QA systems and LLMs are developed and evaluated mainly on Standard Bangla [90], [41], [99], [75]. When a question is written in Chittagonian or Sylheti, or in romanized Bangla, three problems arise. First, dialect words, verb forms and spellings may be misunderstood, and because the question shares few words with Standard Bangla documents, retrieval may fail as well [103], [86]. Second, without grounding in domain-appropriate evidence, a general model may answer health and education questions from memory, which risks hallucinated or unsafe answers [53], [104]. Third, the performance gap between Standard Bangla users and dialect users is not measured, because existing benchmarks do not report results by language variety.

Formally, a user question q is written in a variety $v \in \{\text{Standard, Chittagonian, Sylheti}\}$ (in Bangla or Latin script) and belongs to a domain $d \in \{\text{education, general knowledge, medical, out-of-scope}\}$. The system must identify v and d, produce a Standard Bangla version of q when v is a dialect or the text is romanized, retrieve evidence from the knowledge base of d, and return a Standard Bangla answer grounded in that evidence; or it must ask for clarification, abstain, or give an emergency referral when an answer would not be reliable or safe. Answer quality should be as similar as possible across the three varieties.

This thesis answers the following research questions:

> **RQ1** How accurately, and with how well-calibrated confidence, can a lightweight router identify the language variety and the domain of a Bangla question?
>
> **RQ2** How well can a parameter-efficiently fine-tuned model normalize Chittagonian, Sylheti and romanized questions into Standard Bangla, and how much of the dialect gap in retrieval and answer quality does this normalization close?
>
> **RQ3** How much do retrieval, fine-tuned domain experts, domain routing and the answer/clarify/abstain controller contribute to answer quality and behaviour, and do routed experts outperform a single shared adapter?
>
> **RQ4** Which open base model offers the best balance of answer quality, dialect handling and compute cost under an identical fine-tuning budget, and does Bangla-specific pre-training help after fine-tuning?

1.4 Objectives

The main objective is to design, implement and evaluate a dialect-aware, multi-domain Bangla QA framework that accepts questions in Standard Bangla, Chittagonian or Sylheti, in Bangla or Latin script, and returns grounded Standard Bangla answers for education, general-knowledge and medical questions. The main deliverables are trained routing models, a dialect normalization expert, domain-specific RAG experts, an evaluation suite that reports every result separately for each language variety, and a reproducible code base. The specific objectives are:

- to build lightweight, calibrated routers that identify the language variety and the domain of a question [51], [40];
- to train a dialect normalization expert that rewrites Chittagonian, Sylheti and romanized questions into Standard Bangla with parameter-efficient fine-tuning [103], [43];
- to build a separate knowledge base and retriever for each domain, and to adapt one shared open base model to each domain with LoRA/QLoRA experts [33], [50];
- to decide, for every question, whether to answer, ask for clarification or abstain, and to handle medical emergencies and medicine-dose requests with deterministic safety rules and a doctor-referral note [24], [30];
- to compare seven open base models, including Gemma 4 and two Bangla-adapted models (TigerLLM and TituLM) with their original versions, under an identical fine-tuning budget [90], [88], [102];
- to evaluate every component and the complete system, using NCTB-QA, BanglaRQA and dialect resources, with results reported separately for Standard Bangla, Chittagonian and Sylheti, with ablations, confidence intervals and significance tests, and to test the system on dialect questions written by native speakers [99], [41].

1.5 Methodology in Brief

The framework follows a *route, normalize, then ground* design (Figure 4.1 in Chapter 4). A question first passes rule-based safety checks for emergencies and dose requests. A calibrated linear classifier on character and word n-grams identifies the variety; if the question is Chittagonian or Sylheti with enough confidence, or romanized, a normalization expert rewrites it into Standard Bangla. A second classifier chooses the domain, and a controller decides whether to answer, ask for clarification or abstain. For answerable questions, the retriever of the chosen domain searches that domain's knowledge base (BM25 for education and general knowledge, dense BGE-M3 retrieval for medical questions [14], [62]), and the domain's LoRA expert writes the answer from the three best passages. All experts are small adapters on one 4-bit Gemma-4-E4B-it model [102], a *Mixture-of-Adapters* [43], [25], [18]. An output

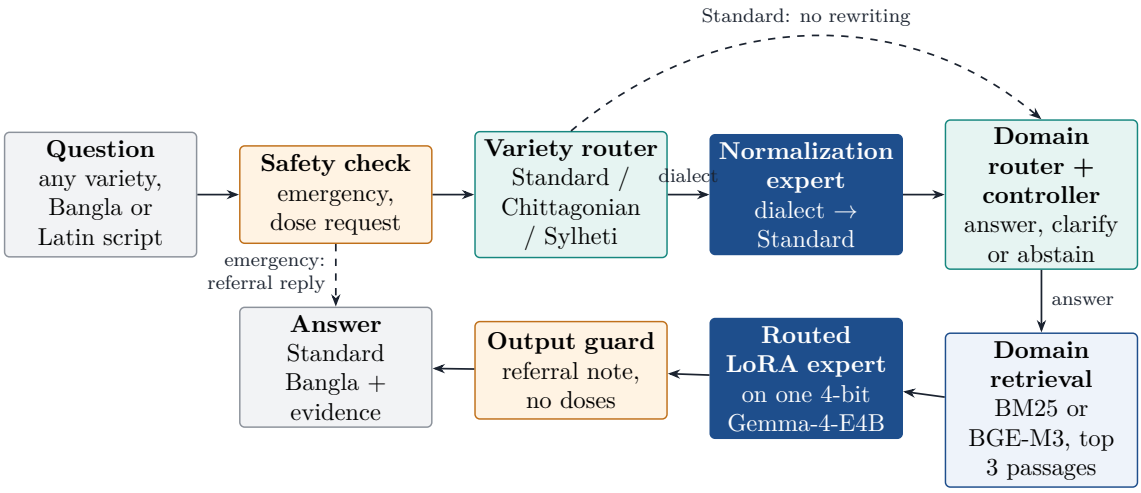


Figure 1.2: Overview of the proposed *route, normalize, then ground* framework. The routers and the controller run on the CPU; the normalization expert and the domain experts are LoRA adapters on one shared 4-bit model (a Mixture-of-Adapters). Each component is described in Chapter 4 (Figure 4.1).

guard adds a doctor-referral note to medical answers and refuses to give medicine doses. Figure 1.2 gives an overview of this pipeline.

The work was carried out in stages. The data stage cleaned and split more than twenty public data sources, including four dialect corpora (Vashantor, Kothon, ChatgaiyyaAlap and ONUBAD), two QA datasets with unanswerable questions (NCTB-QA and BanglaRQA), textbooks and medical resources [51], [100], [82], [93], [99], [41], [56], [97]. Seven open models, including the Bangla-adapted TigerLLM and TituLM and the multilingual Gemma-4-E4B, Qwen3 and Llama-3.2, were then compared under an identical QLoRA budget [90], [88], [102], [95], [72], [50]. The experts, routers and retrievers were trained and chosen on validation data, and the test data were used only once, for reporting. Finally, the complete system was evaluated under nine conditions on 2,225 paired test questions (780 Standard questions with 720 Chittagonian and 725 Sylheti versions), and on 481 dialect questions written by native speakers. Every result is reported separately for each language variety.

1.6 Contributions

1. **A modular dialect-aware QA framework** for Bangla that separates variety routing, dialect normalization, domain routing, answer/clarify/abstain control, per-domain retrieval, routed LoRA experts on one shared base model, and medical safety rules, together with its complete, reproducible implementation.

2. **A fine-tuned dialect normalization expert** for Chittagonian, Sylheti and romanized input that raises chrF++ from 51.2 to 76.3 (Chittagonian) and from 55.4 to 74.0 (Sylheti) after one epoch (a second epoch adds 1.3 chrF++ overall), transfers to three dialects never seen in training, beats similarity-based few-shot prompting, and on the official Vashantor test sentences equals

or exceeds published dedicated translation models except one (Chittagonian BLEU against DialectBanglaT5).

3. **A paired, per-variety evaluation protocol** (the same questions in three varieties, nine system conditions, Holm-corrected significance tests) that measures the dialect gap end to end. It shows that normalization closes 61% (Chittagonian) and 64% (Sylheti) of the gap without harming Standard Bangla input; to our knowledge, no earlier Bangla work measured this downstream effect.

4. **An evaluation on 481 dialect questions written by native speakers**, which shows that with normalization the system treats a dialect question like its Standard version 86.1% of the time (65.3% without).

5. **A controlled model comparison and informative negative results**: under an identical budget, Gemma-4-E4B-it is the best of seven open models; Bangla continued pre-training gives no advantage after fine-tuning; routed experts equal a single pooled adapter in accuracy, so their value is modularity; and a contrastively trained dialect-invariant retriever is not better overall than its base encoder and does not reduce the dialect gap.

1.7 Scope and Challenges

This thesis builds and evaluates a research prototype, not a ready-to-use medical product; both its individual parts and the complete system are evaluated. Its scope and main challenges are listed below.

Scope of the Study

- **Input:** text questions in Standard Bangla, Chittagonian or Sylheti, written in Bangla script or romanized. Three further dialects (Barishal, Mymensingh, Noakhali) are used only as unseen test data. Speech input is not part of this study, although a Bangla medical speech recogniser with Sylheti recordings [69] could be placed in front of the system later.

- **Domains:** education (NCTB curriculum), general knowledge (Bangla Wikipedia passages of BanglaRQA) and medical questions, each with its own knowledge base and retrieval pipeline; other questions are recognised and declined.

- **Output:** answers in Standard Bangla, with the evidence shown to the user, because the knowledge sources and most readers use the standard form. Answering back in the user's dialect is left for future work.

- **Healthcare:** general health information only, with emergency referral and a doctor-referral note; the system does not diagnose or prescribe.

- **Compute:** open models trained with QLoRA on one rented NVIDIA A40 GPU, with all non-GPU work on free Google Colab CPU runtimes [43], [50].

Key Challenges

- **Scarce and uneven dialect data:** Parallel dialect data are limited and uneven across dialects, and most of them are everyday conversation rather than health or school questions [51], [86], [100], so domain vocabulary such as symptom names is under-represented. For example, the medical dialect extension of BanglaCHQ-Summ contains 2,350 Sylheti but only 500 Chittagonian questions [86].
- **No standard dialect spelling:** Chittagonian and Sylheti have no agreed written form, so the same word appears with several spellings, and romanized text shares almost no words with Bangla-script text [103].
- **No dialect QA test data:** No existing dataset contains health or education questions in Chittagonian or Sylheti with reference answers, so the dialect gap has to be measured with generated paired questions and with human-written dialect questions without reference answers.
- **Error propagation:** A wrong variety or domain decision, or a poor normalization, affects every later stage; the system therefore uses calibrated confidences and a controller rather than trusting each component blindly.
- **Medical safety:** Wrong, overconfident or delayed health answers can cause harm, and triage is hard even for large models [104]; emergencies and dose requests must be handled deterministically.
- **Unanswerable and vague questions:** Many questions cannot be answered from the knowledge base or are under-specified; the system must abstain or ask instead of guessing [24], [30], [99].
- **Evaluating free-text answers:** Medical answers are free text, and word-overlap metrics under-rate correct answers that use different words; several metrics, a check of grounding and human review are needed.
- **Limited compute and budget:** All training had to fit one rented GPU and a small budget (about 40 GPU-hours, roughly US$20), which makes quantized, parameter-efficient training essential [50].

Chapter 2

Literature Review

The work relevant to this thesis comes from research streams that have developed largely separately: Bangla dialect processing, Bangla question answering, Bangla language models, and applications of language technology in healthcare and education. This chapter first introduces the concepts the thesis builds on, then reviews the related work, highlighting its achievements and limitations, and ends with the gaps that the proposed framework addresses.

2.1 Preliminaries

2.1.1 Bangla and Its Regional Varieties

Standard Bangla (cholito bhasha) is the form used in formal writing, education, broadcasting and administration. The regional varieties form a continuum across Bangladesh and differ from the standard in pronunciation, vocabulary, verb morphology and sometimes word order, in some regions vastly so [103], [51]. Chittagonian, spoken in the south-east, and Sylheti, spoken in the north-east, are among the most divergent. Online, dialect speakers write in Bangla script with phonetic spellings of dialect words, or in Latin script (romanized Bangla, often called Banglish); Sylheti also has a historical script of its own, Sylheti Nagri [103]. A dialect sentence and its Standard equivalent can share very few word forms, sometimes none, although they convey exactly the same meaning, for example Sylheti *kita* for Standard *ki* ("what"), which is why systems trained on Standard Bangla misread dialect input.

2.1.2 Dialect Identification and Normalization

Dialect identification predicts the variety a text is written in and is usually treated as text classification [51]. Dialect normalization rewrites a dialect text in the standard variety while keeping its meaning, which makes it a form of machine translation [51], [103]. Its quality is measured with BLEU [8], the character n-gram F-scores chrF and chrF++ [15], [17], and word or character error rate (WER, CER). A useful sanity check is the copy-input baseline, which scores the unchanged dialect input against the Standard reference: a system that scores below it has made the text worse [103].

Both tasks are long-standing problems in NLP for closely related languages, varieties and dialects [35]. Dialect identification is usually framed as supervised text classification, and in shared tasks for Arabic it has reached high scores: the best systems of NADI 2023 reached a macro-F1 of 87.27 on country-level Arabic dialect identification [48]. Normalization is much harder than identification. In the same shared task, the best systems that translated Arabic dialects into Modern Standard Arabic reached only 14.76 and 21.10 BLEU on the two machine-translation subtasks [48]. Normalization is easier to learn when parallel dialect–standard data exist, as in the MADAR corpus of 25 city dialects [20], and character-level models suit it because dialect words are often spelling variants of standard words: for Swiss German, character-based neural translation was the best normalization strategy [21].

Romanized input is a related problem. Bangla written in Latin letters has no fixed spelling, so the same word appears in many forms, and converting it back into Bangla script (back-transliteration) is a task of its own. BanglaTLit provides 42.7k romanized–Bangla-script pairs for this task, together with a 245.7k-sample romanized pre-training corpus [64]. In this thesis, romanized questions are normalized by the same expert that normalizes dialect questions, so back-transliteration and dialect normalization are handled in one step.

2.1.3 Question Answering

In extractive QA the answer is a span of a given passage, scored with exact match (EM) and token-level F1 [41]; in multiple-choice QA the system picks an option and is scored by accuracy [75], [92]; in open-domain QA the system must first find the evidence itself, usually with a retriever followed by a reader [31]. When the answer is generated in the system's own words, evaluation relies on semantic similarity measures such as BERTScore [36] and on checking whether the generated answer is supported by the evidence [63]. A realistic system must also recognise questions it cannot answer. SQuAD 2.0 added more than 50,000 unanswerable questions written adversarially by crowd workers to look answerable, and showed that models trained only on answerable questions answer them confidently [24]. Selective QA trains a calibrator that lets a model abstain when it is unsure, trading coverage for accuracy [30].

Most of the early Bangla QA systems follow the reading-comprehension setting, where the passage is given for context and answer searching. A practical system must instead first retrieve relevant documents from a knowledge base with a retriever and then pass the top-k documents to a reader model, which is the retriever–reader design of open-domain QA [31]. The framework in this research follows that design, with an LLM as the reader.

2.1.4 Large Language Models and Parameter-Efficient Fine-Tuning

Modern LLMs are Transformers pre-trained on large text collections and then instruction-tuned. Updating all their parameters is too expensive for most academic budgets. LoRA freezes each pretrained weight matrix $W \in \mathbb{R}^{d \times k}$ and learns a

low-rank update $\Delta W = BA$ with $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$ and $r \ll \min(d, k)$ [43]. Only A and B are trained, which reduces the number of trainable parameters by several orders of magnitude, and different adapters can be attached to the same base model for different tasks. QLoRA stores the frozen base model in 4-bit NormalFloat (NF4) precision with double quantization and paged optimizers and back-propagates through it into the LoRA adapters, so billion-parameter models can be fine-tuned on one GPU with little loss in quality [50].

2.1.5 Retrieval-Augmented Generation

RAG combines a retriever that finds passages relevant to the question with a generator that writes the answer from them [33]. Sparse retrieval such as BM25 scores documents by weighted term overlap and works well when question and document share words [14]. Dense retrieval encodes questions and passages as vectors and ranks by vector similarity, which also matches paraphrases and synonyms [31]; multilingual embedding models such as multilingual E5 and BGE-M3 support Bangla [62], [78]. Hybrid systems fuse sparse and dense rankings, typically with reciprocal rank fusion (RRF) [13]. Self-RAG trains the model to decide when to retrieve and to critique both the retrieved passages and its own output [59], and Adaptive-RAG uses a small classifier to choose between no retrieval, single-step and multi-step retrieval according to question complexity [67].

2.1.6 Mixture of Experts and Routing

A mixture of experts is a set of specialised models with a gating function that decides how much each model contributes to each input [3]. In current LLMs, sparse mixture-of-experts layers send each token to a few expert sub-networks inside one model, which increases capacity at little extra compute [18], [42]. This thesis applies the idea at the system level rather than inside the network: separate lightweight routers send each question to one normalization expert and one domain expert, and each expert is an independently trained LoRA adapter on a shared base model. This is closer to Adaptive-RAG's small routing classifier [67] and to adapter-based transfer [25] than to a sparse MoE layer, so we call it a *Mixture-of-Adapters*. It keeps each expert small, interpretable and replaceable. Related methods that combine several adapters are reviewed in Section 2.2.7.

2.2 Review of Existing Research

Research relevant to dialect-aware Bangla QA has expanded rapidly since 2022. The reviewed work falls into nine themes: dialect resources and translation; dialects in health and LLM prompting; Bangla QA benchmarks; Bangla-capable language models; Bangla healthcare and education applications; dialect processing beyond Bangla; combining and routing LoRA adapters; re-ranking and hybrid retrieval; and grounding, abstention and dialect robustness. Table 2.1 summarises the most closely related studies.

2.2.1 Dialect Resources and Dialect–Standard Translation

Vashantor was the first large benchmark for translating Bangla regional dialects into Standard Bangla [51]. It is multilingual (Bangla, romanized Bangla and English) and covers five regions (Chittagong, Noakhali, Sylhet, Barishal and Mymensingh) with 32,500 sentences: 2,500 Standard sentences rendered in every region give 12,500 dialect–Standard pairs, with an official split of 1,875 training, 250 validation and 375 test sentences per region. In the first version of the paper, fine-tuned BanglaT5 reached a dialect→Standard BLEU of 44.03 for Chittagong and 51.08 for Sylhet (mT5: 36.75 and 51.32), and region detection reached 85.86% accuracy, with Sylhet the hardest region (F1 0.75) [51]. The revised version introduces DialectBanglaT5 and DialectBanglaBERT, reporting a best BLEU of 71.93 and lowest WER of 0.1470 (Mymensingh) and 89.02% region-detection accuracy (Chittagong F1 0.9241). These results show that fine-tuning works across dialects, but the data are short everyday sentences, not domain-specific, and translation is the end task.

Several newer corpora extend the dialect data. Kothon provides human-validated Chittagonian–Standard and Sylheti–Standard sentence pairs [100], ONUBAD covers several regional dialects [93], and ChatgaiyyaAlap provides Chittagonian–Standard pairs [82]. These corpora supply most of the training pairs of our normalization expert. 5-Dialects-BN contains 6,000 manually verified utterances in five dialects (1,900 Chittagong, 1,500 Noakhali, 1,200 Sylhet, 700 Barisal, 700 Rangpur), each with a romanized transliteration, a Standard Bangla translation, an English translation and a subjectivity label; Cohen's κ for the Standard Bangla renderings was 0.74 [103]. It evaluated seven LLMs with zero-shot, few-shot and chain-of-thought prompting and with LoRA. LoRA ($r = 16$, $\alpha = 32$) on 160 examples per dialect raised Mistral-7B from 16.2 to 73.6 BLEU on dialect→English translation, above the zero-shot score of Gemini 3 Flash (46.4). Dialect→Standard normalization was evaluated only with prompting: some open models scored below the copy-input baseline (39.66 chrF++), a fine-tuned BanglaT5 reached 44.47 chrF++, and the best system, few-shot Gemini, reached 78.70 chrF++. All models degraded on romanized input.

Work in the opposite direction translates Standard Bangla into dialects. Bridging Dialects fine-tuned BanglaT5, mT5 and mBART-50 to translate Standard Bangla into the five Vashantor dialects; after 50 epochs BanglaT5 reached a WER of 0.2380 and CER of 0.1216 on Mymensingh but 0.7066 and 0.3706 on Chittagong, with Chittagong and Sylhet the hardest [70]. BanglaDialecto built an end-to-end speech standardization pipeline for Noakhali: about 10 hours of speech, a fine-tuned Whisper recogniser (CER 0.8%, WER 1.5%), BanglaT5 translation into Standard Bangla (BLEU 41.6) and speech synthesis [74]. It shows that dialect standardization can serve as an interface for other systems, but it covers one dialect and stops before any downstream task. None of these works connects dialect translation to retrieval or to answering a user's question.

2.2.2 Dialects in Health and LLM Prompting

BanglaCHQ-Prantik translated the 2,350 consumer health questions of BanglaCHQ-Summ into Sylheti and 500 of them into Chittagonian with native-speaker transla-

tors, creating the first medical-domain dialect benchmark [86]. The smaller Chittagonian subset reflects the fact that few annotators can read and write Chittagonian and that it has no standard orthography. It evaluated Qwen 2.5 3B, Gemma 3 1B, GPT-4o-mini and Gemini 2.5 Flash with zero-shot, one-shot, five-shot and chain-of-thought prompting in the Standard→dialect direction; the best BLEU was 23.67 (Sylheti, Gemini, five-shot) against 14.78 for Chittagonian, with frequent errors in medical terms, numerals, idioms and spelling. This benchmark contains exactly the kind of dialectal health questions a healthcare QA system must handle, and it shows that prompting frontier models is not enough. Sylheti-CAP studied Sylheti translation in both directions on the Vashantor test set with context-aware prompting of several LLMs, reaching at most 53.72 chrF for Sylheti→Standard [89]. A comparison of RAG pipelines for Standard→dialect translation in six dialects found that ten LLMs reached only 5–12 BLEU (WER 67–84%) without retrieved examples, and that retrieving aligned sentence pairs worked better than retrieving transcripts (Chittagong BLEU 9→26, WER 76%→55%) [91]. A smaller model, Llama-3.1-8B (WER 51.18), even outperformed the much larger GPT-OSS-120B (WER 52.65) [91], which indicates that parallel examples matter more than model size. The same group benchmarked 19 LLMs on questions in nine Bengali dialects and found far lower quality for Chittagonian (5.44/10) than for Tangail (7.68/10) [106]. AmarDoctor, a multilingual triage application (81.08% top-1 diagnostic precision on 185 clinical vignettes), took a lexicon approach: its medical team linked 908 symptoms to 3,237 local variants, including Chittagonian and Sylheti, and the authors report that a general speech-to-text service struggled with dialect speech because of phonetic differences [87]. A lexicon, however, only covers the terms it contains. Together, these works show that prompting alone is weak for dialects, that retrieved examples help, and that medical dialect data are scarce, especially for Chittagonian.

2.2.3 Bangla Question Answering Benchmarks

BanglaRQA is a human-annotated reading-comprehension dataset of 14,889 question–answer pairs on 3,000 Bangla Wikipedia passages, 3,631 of them unanswerable, with four question types (factoid, causal, confirmation and list) and three answer types (single span, multiple spans and yes/no); fine-tuned BanglaT5 reached 62.42 EM and 78.11 F1, with weaker performance on the more difficult question types [41]. The passages cover a wide variety of general-knowledge topics.

NCTB-QA contains 87,805 question–answer pairs from 50 national-curriculum textbooks (grades 1–10), 57.25% answerable and 42.75% unanswerable with plausible distractors, split 80/10/10 so that no context appears in two splits; fine-tuning raised BERT from 0.150 to 0.620 F1 [99]. The dataset is closely aligned with the national curriculum, which makes it a natural basis for the education expert; however, its questions were model-generated from the textbook contexts, so real student question types may be under-represented.

BEnQA has 5,161 parallel Bangla–English science multiple-choice questions from grade 8, 10 and 12 board exams; LLMs were consistently less accurate in Bangla, chain-of-thought helped reasoning questions, and appending the English translation of the question helped in all subjects [75]. SOMAJGYAAN has 4,234 multiple-choice questions on Bangladeshi history, geography, economics, law and society, extracted

from national-curriculum books, where closed models led (Gemini 2.0 Flash 0.73, GPT-4o 0.69) and LoRA fine-tuning of open models did not close the gap [92].

Earlier systems were extractive and assumed that the correct passage is given. Mayeesha et al. translated SQuAD 2.0 into Bangla, fine-tuned multilingual BERT, DistilBERT and RoBERTa, tested them on a small human-annotated set from Bangla Wikipedia, and compared the model answers with answers from Bangla-speaking children as a human reference point [37]. Khondoker et al. built about 3,000 question–answer pairs from NCTB textbooks of grades 6–10, where Bangla-BERT reached F1 0.75 and EM 0.53 and RoBERTa Base performed worst (F1 0.19); dataset size and spelling errors were the main constraints [71].

BanglaCQA extends BanglaRQA with 6.3K counterfactual contexts (42.28% more contexts than the original) to test whether models use the passage or their memory; under counterfactual passages, models largely followed the passage and could not recover memorised answers (parametric similarity about 0.13–0.14) [98]. Combining factual, counterfactual and answerability data improved contextual answering, chain-of-thought prompting helped decoder-only LLMs distinguish factual from counterfactual knowledge, and the authors found LLM-based semantic similarity more reliable than traditional overlap scores for comparing Bangla answers [98]. For a RAG system, this means that the generator has to be trained to rely on the retrieved evidence. All of these benchmarks are in Standard Bangla, and none contains dialect questions.

2.2.4 Bangla-Capable Language Models

BanglaBERT and BanglaT5 remain strong and efficient Bangla baselines for classification and generation [40], [49]. TigerLLM continually pre-trains small models on about 10 million tokens from 163 NCTB textbooks (grades 6–12) and instruction-tunes them on 100,000 Bangla instructions created from 500 human-written seed tasks; the paper describes 1B (LLaMA 3.2) and 9B (Gemma 2) models and reports that they outperform other open Bangla LLMs [90]. The released 1B checkpoint, however, has exactly the architecture and tokenizer of Gemma-3-1B, which is why this thesis pairs it with Gemma-3-1B. TituLLMs continually pre-train Llama-3.2 models on about 37 billion Bangla tokens with an extended tokenizer; the 3B model is a base (not instruction-tuned) model [88]. General multilingual open models are the alternative: Gemma 3 and Gemma 4 [102], [83], Qwen3 [95], Llama 3.2 [72] and Tiny Aya, whose "Fire" variant targets South Asian languages [105]. Gemma 4 is trained on over 140 languages, is available in small "effective 2B" and "effective 4B" edge sizes with a context of up to 128K tokens, and is licensed under Apache 2.0 [102]. Whether Bangla continued pre-training still helps once a model is fine-tuned for the task had not been tested under a controlled budget; Chapter 5 does so.

Evaluations of general LLMs on Bangla point the same way. BenLLM-Eval tested ChatGPT, LLaMA-2 and Claude-2 zero-shot on a range of Bengali NLP tasks. In a few tasks these models matched or exceeded fine-tuned models, but in most tasks their performance was poor [68]. Massively multilingual instruction tuning is one response: the Aya model follows instructions in 101 languages, more than half of them lower-resourced, and outperforms earlier multilingual instruction-tuned models such as mT0 and BLOOMZ on most tasks [77]. On Bangla QA, LLMs still answer

less accurately than in English [75]. Together, these results suggest that an open multilingual model needs task-specific adaptation for Bangla. They also motivate comparing several open base models under one identical fine-tuning budget, rather than choosing one by reputation.

2.2.5 Bangla Healthcare and Education Applications

DocTalkBN is a corpus of expert telemedicine conversations from national health programmes: 557.63 hours of audio, 1,515 multi-turn patient conversations and 10,274 host–doctor question–answer transcripts in 26 specialties, about 1.7 million tokens in total [104]. Its benchmarks show that classifying doctor advice as safe or harmful is easy for current models (macro-F1 above 0.92), while four-way triage is hard (best macro-F1 0.463); the authors name retrieval-augmented generation over the corpus as future work. BanglaCHQ-Summ contains 2,350 real consumer health questions from an online consultation platform with human-written summaries; fine-tuned BanglaT5 reached a ROUGE-L of 48.35 [55]. BanglaMedQA evaluates RAG strategies for Bangla medical multiple-choice questions and finds large gains from retrieval (for example 42.15% $\rightarrow$ 66.60% for Llama-3.1-8B) [94], in line with English evidence that retrieval improves medical QA [80] and that LLMs encode clinical knowledge [57].

Medical entity recognition on such informal text is hard. Bangla-HealthNER annotates 31,783 consumer health texts from a public online health platform with seven entity types (symptom, health condition, age, medicine, dosage, medical procedure and specialist), and the best model, BanglishBERT, reaches an F1 of only 56.13% [56]. The texts frequently switch between Bangla and English and come from writers of different regions, and Bangla's rich inflection and flexible word order make entity extraction harder than in English. Bangla MedER annotates 6,895 medical statements with entities such as medicine and chemical names, diseases, organs, pharmacological classes and hormones, and reports 89.58% accuracy with an ensemble of two BERT branches, 11.80 percentage points above a single BERT model [97].

For education, a cross-national survey of 318 university students from the US and Bangladesh found that students already use LLMs for learning, with LLM use correlated with perceived literacy benefits ($r = .59$) and optimism towards AI ($r = .41$), alongside concerns about over-reliance, ethics and critical thinking [84]. This evidence, however, concerns university students who are comfortable with English and Standard Bangla, not school students who might benefit most from curriculum support in their own dialect. Farzana et al. addressed the grading of written answers, which is still done manually in remote and rural schools with few qualified teachers: they built a synthetic bilingual dataset of 122,710 questions with 490,391 question–answer instances and fine-tuned Qwen3-8B with QLoRA to produce a score and short feedback, which graded Bangla and English written answers close to teachers (Spearman $\rho = 0.936$, MAE $= 0.725$ on 259 teacher-graded instances) [101]. This shows that parameter-efficient fine-tuning of open models is practical for Bangla education, and that answers should be judged by meaning rather than word overlap. None of these systems accepts dialect questions, retrieves evidence and generates grounded answers.

2.2.6 Dialect Processing Beyond Bangla

The dialect problem is not specific to Bangla. DIALECTBENCH collected 10 text-level tasks covering 281 varieties and found clear performance gaps between standard and non-standard varieties of the same language [65]. A study of LLMs on regional dialects in machine translation and speech recognition found that the size of this dialect gap depends on the training data, but not consistently across models and languages, so that no single fix works everywhere [54]. Arabic, which has a standard written variety and many spoken dialects, is the best-studied case: the MADAR corpus provides parallel sentences in 25 city dialects [20], and the NADI shared tasks have run dialect identification at country and province level (NADI 2020 covered 100 provinces in 21 countries) [28] and, more recently, dialect-to-standard translation, where scores are still low [48]. Swiss German, which is spoken but rarely written in a standard form, has been normalized into Standard German before translation with about 60,000 parallel words, and character-based translation worked best [21].

Two lessons carry over to Bangla. First, the dialect gap must be measured, because benchmarks in the standard variety hide it [65], [54]. Second, normalizing dialect input into the standard variety lets existing standard-variety systems be reused [21]. Most of this work, however, evaluates identification or translation as end tasks. This thesis measures the effect of normalization on a downstream task, question answering, using the same questions in each variety.

2.2.7 Combining and Routing LoRA Adapters

Adapters are small trainable modules added to a frozen pre-trained model [25], and LoRA is a low-rank form of them [43]. Because each adapter is small, one base model can hold many of them, which raises the question of how to combine them. AdapterFusion learns to combine the representations of several task adapters for a new target task and outperformed full fine-tuning and multi-task learning on 16 language-understanding tasks [38]. LoraHub composes several LoRA modules for an unseen task by searching for mixing weights from a few examples, and comes close to in-context learning without needing the examples in the prompt [66]. Mixture of LoRA Experts (MoLE) learns a gating function that weights the outputs of several LoRA modules in each layer, and fuses them better than simple arithmetic merging [79]. X-LoRA uses the model's hidden states to mix LoRA experts at every layer and token [61]. On the systems side, S-LoRA serves thousands of LoRA adapters from one base model, with up to four times the throughput of earlier serving libraries [76], which shows that many adapters on one shared model is a practical way to deploy several skills.

These methods mix adapters *inside* the network, with weights that are learned and that change from token to token. This thesis takes a simpler, system-level approach: a separate, inspectable CPU router chooses *one* whole adapter per step (the normalization expert, then one domain expert), so every decision can be logged, evaluated and corrected on its own. Because a learned mixture would be harder to inspect and to test component by component, the thesis instead compares routed experts with a single pooled adapter trained at equal compute.

2.2.8 Re-ranking and Hybrid Retrieval

A retriever such as BM25 [14] or a dense encoder [31], [62] returns candidates quickly, but its ranking is often imperfect. Re-ranking passes the top candidates to a stronger model that reads the question and each passage together. A BERT cross-encoder used in this way greatly improved passage ranking [27], and the sequence-to-sequence ranker monoT5 was especially strong when training data were scarce and transferred zero-shot to other collections [34]. ColBERT keeps much of this quality at far lower cost by comparing token embeddings late in the computation, and runs two orders of magnitude faster than BERT-based rankers [32]. Multilingual cross-encoders such as bge-reranker-v2-m3 make re-ranking available for Bangla [60]. Rankings from several retrievers or several query formulations can also be fused with Reciprocal Rank Fusion [13], as in RAG-Fusion [73]. When the right collection is uncertain, a router can choose between retrievers, as RouterRetriever does over a mixture of expert embedding models [85]. Bangla QA work has used retrieval mainly with a single retriever [94]. This thesis chooses a retriever per domain on validation data, and its answer-time extension adds cross-encoder re-ranking and a search of a second knowledge base when the domain router is unsure (Chapter 4).

2.2.9 Grounding, Abstention and Dialect Robustness

Grounding answers in retrieved evidence is the standard remedy for hallucination in knowledge-intensive tasks [53], [33]. Self-RAG and Adaptive-RAG decide when and how to retrieve [67], [59], SQuAD 2.0 and selective QA teach models to abstain [24], [30], and BanglaRQA and NCTB-QA provide real unanswerable Bangla questions for learning this behaviour [41], [99]. For dialect robustness in English, Multi-VALUE generates synthetic dialect variants for evaluation and augmentation [58], and TADA trains a task-agnostic dialect adapter that aligns dialect and standard representations; its gains are larger for dialects far from the standard (+2.8 to +3.9 points) than for dialects close to it (+0.3 to +0.4) [52]. What is missing is a system that treats uncertainty about the question itself, caused by dialect and routing errors, as well as uncertainty about the evidence.

Table 2.1: Summary of the most closely related studies and the gap each leaves for this thesis.

Work	**Task and data**	**Method**	**Key result**	**Gap**
Vashantor [51]	Dialect→ Standard translation and region detection; 5 regions, 12,500 pairs	BanglaT5, mT5; Dialect-BanglaT5/ BERT (v2)	BLEU 44.03 (Ctg), 51.08 (Syl) with BanglaT5; 89.02% region accuracy (v2)	Everyday sentences; no QA

Work	Task and data	Method	Key result	Gap
Kothon, ONUBAD, ChatgaiyyaAlap [100], [93], [82]	Dialect–Standard parallel corpora	Dataset papers	Human-validated pairs	No downstream task
5-Dialects-BN [103]	6,000 utterances, 5 dialects, 5 aligned fields	7 LLMs; prompting and LoRA	LoRA (160/ dialect) 73.6 BLEU to English; some models below copy baseline	Normalization only prompted; no QA
Bridging Dialects [70]	Standard→ dialect on Vashantor	BanglaT5, mT5, mBART-50	WER 0.71 (Ctg) vs 0.24 (Mym)	Reverse direction; no QA
BanglaDialecto [74]	Noakhali speech standardization, 10 h	Whisper, BanglaT5, TTS	ASR CER 0.8%; BLEU 41.6	One dialect; no QA
Bangla CHQ-Prantik [86]	Health questions: 2,350 Sylheti, 500 Chittagonian	4 LLMs, prompting	Best BLEU 23.67 (Syl, 5-shot)	Standard→ dialect; no fine-tuning; no QA
Sylheti-CAP [89]	Sylheti→ Standard, Vashantor test	Context-aware LLM prompting	Best chrF 53.72	Prompting only; no QA
RAG dialect MT [91]	Standard→ dialect, 6 dialects	RAG few-shot, 10 LLMs	BLEU 5–12 without examples; Chittagong 9→ 26 with sentence-pair retrieval	Translation only
BanglaRQA [41]	Reading comprehension, 14,889 QA, 3,631 unanswerable	BanglaT5 and encoders	62.42 EM, 78.11 F1	Standard only; passage given
NCTB-QA [99]	Textbook QA, 87,805 pairs, 42.75% unanswerable	BERT, RoBERTa, ELECTRA	BERT F1 0.150→0.620	Standard only; passage given; model-generated questions
BEnQA, SOMAJ-GYAAN [75], [92]	Bangla MCQ benchmarks	LLM prompting	Bangla below English; closed models lead	Standard only; multiple choice

Work	Task and data	Method	Key result	Gap
bn_squad [37]	SQuAD 2.0 translated to Bangla; small human-annotated Wikipedia QA set	mBERT, DistilBERT, RoBERTa	Established machine-vs-human (children) baseline gap	Translated data is noisy; no retrieval-augmented answering
NCTB Multi-BERT QA [71]	NCTB passage QA, classes 6–10; about 3,000 pairs	BERT Base, Bangla-BERT, RoBERTa	Bangla-BERT F1 0.75, EM 0.53	Passage given; Standard Bangla
BanglaCQA [98]	Counterfactual contexts for BanglaRQA	BanglaT5, LLMs	Models follow the passage	No retrieval or dialects
TigerLLM, TituLLMs [90], [88]	Bangla LLMs	Continued pre-training	Better than other open Bangla LLMs	No dialects; not compared after task fine-tuning
AmarDoctor [87]	Triage app, 185 vignettes	Knowledge graph, dialect symptom lexicon	81.08% top-1 diagnostic precision	Lexicon coverage; not QA
DocTalkBN [104]	Telemedicine corpus, 10,274 QA	LLM and encoder benchmarks	Triage macro-F1 0.463; safety F1 > 0.92	No retrieval; no dialects
Bangla CHQ-Summ [55]	2,350 real consumer health questions with human summaries	Fine-tuned BanglaT5 for summarization	ROUGE-L 48.35%	Summarization only, not QA; no answers provided
Bangla-MedQA [94]	Bangla medical MCQ	6 RAG strategies	42.15% $\rightarrow$ 66.60% with RAG	Standard only; MCQ

2.3 Summary of Findings and Research Gap

- Regional varieties differ strongly from Standard Bangla, and Chittagonian and Sylheti are among the hardest to model in both translation directions [103], [51], [70], [86].
- Parallel dialect data exist (Vashantor, Kothon, ONUBAD, ChatgaiyyaAlap, 5-Dialects-BN, BanglaCHQ-Prantik) and offer several thousand dialect–Standard pairs for each target dialect, but they are mostly everyday sentences, and medical dialect data are scarce, especially for Chittagonian [51], [103], [86].
- Prompting is not enough for dialects: some LLMs fall below the copy-input

baseline on normalization, and prompted Standard→dialect translation of health questions reaches only modest BLEU [103], [86]. Small amounts of supervised adaptation give large gains [103], [51].

- Retrieval helps both dialect handling and domain QA: retrieved dialect–Standard pairs improve LLM dialect translation [91], and retrieval improves medical QA [94], [80].
- Bangla QA benchmarks with unanswerable questions exist (BanglaRQA, NCTB-QA), but all Bangla QA benchmarks, including BEnQA and SOMA-JGYAAN, are in Standard Bangla, and the earlier QA systems assume the correct passage is given [41], [99], [75], [92].
- LLMs are not as good as English on Bangla questions, and open source models are behind closed source models [75], [92].
- Bangla-adapted LLMs exist [90], [88], but their benefit over general multilingual models after task fine-tuning had not been measured under an equal budget.
- Bangla healthcare AI ranges from knowledge-graph triage to telemedicine corpora [87], [104], [55]; triage remains hard for LLMs, and consumer health text is informal and code-mixed [56], so explicit safety rules and referral are needed.
- In Bangladesh, students are already employing LLM's in learning but the studied tools were in Standard Bangla or English and were used by university level learners [84].
- When the passage contradicts pretraining knowledge, Bangla QA models largely follow the passage [98], which supports grounding answers in retrieved evidence. The previous NCTB and Wikipedia QA systems were extractive and assumed that the passage was given [37], [71].
- Parameter-efficient fine-tuning makes it possible to adapt one open base model to several tasks with small adapters on one GPU [43], [50]. This enables a modular architecture in which the same base model serves as the dialect expert and as the domain experts.
- Dialect gaps are found across languages, and normalizing into the standard variety is an established strategy, but its effect on a downstream task such as question answering is rarely measured [65], [54], [21].
- Many LoRA adapters can share one base model and be served together [76]. Existing methods mix them inside the network with learned weights [79], [66], while explicit, inspectable routing between whole adapters has received less attention.

Hence, this thesis topic, Dialect-Aware Bangla Question Answering with Routed Domain Experts and Retrieval-Augmented Generation for Healthcare and Education, targets three gaps which are yet to be resolved in previous studies, which are outlined below:

- **Dialect-aware QA.** Dialect identification and normalization have been studied as end tasks, but not connected to retrieval and answering: no system lets a Chittagonian or Sylheti speaker ask a health or education question and receive a grounded answer.
- **Grounded and safe answering in one pipeline.** Bangla QA and LLM studies have not combined domain routing, retrieval, abstention, clarification and medical safety rules in one evaluated system.
- **Per-variety evaluation.** Existing Bangla QA results are reported only for Standard Bangla; the end-to-end gap between Standard and dialect users, and how much of it each component closes, has not been measured.

Chapter 3

Requirements, Impacts and Constraints

This chapter fixes what the system must do, the resources it needs, and the societal, environmental, ethical, economic and practical constraints under which it was designed and evaluated. The framework answers questions written in Standard Bangla, Chittagonian or Sylheti, in Bangla script or romanized form, about three domains (school education, general knowledge and healthcare), and always responds in Standard Bangla.

3.1 Final Specifications and Requirements

The final system routes each question by language variety and domain, normalizes dialectal and romanized questions into Standard Bangla, and answers them with domain-specific retrieval-augmented generation. It is designed to give grounded Standard Bangla answers to users of Standard Bangla, Chittagonian and Sylheti, and to ask for clarification, abstain or refer the user to a doctor when an answer would not be reliable.

3.1.1 Functional Requirements

The final system satisfies the following functional requirements (FR).

Table 3.1: Functional requirements, implementing components and verifying experiments.

ID	Requirement	Implemented by	Verified by
FR1	Accept questions in Standard Bangla, Chittagonian and Sylheti, in Bangla script or romanized	script detection + dialect router	router test set, end-to-end test

ID	Requirement	Implemented by	Verified by
FR2	Identify the language variety of each question	calibrated TF-IDF dialect router	per-class F1, unseen dialects
FR3	Rewrite dialectal or romanized questions into Standard Bangla, preserving symptoms, numbers and named entities	LoRA normalization expert	chrF++/BLEU on held-out pairs
FR4	Decide the domain (education, general, medical, out-of-scope)	domain router	per-class F1, end-to-end domain accuracy
FR5	Ground every answer in retrieved evidence from the domain knowledge base	per-domain retriever (BM25 or dense)	hit@k, MRR, nDCG@10
FR6	Answer with a domain-specific expert on one shared base model	Mixture-of-Adapters (LoRA experts)	F1 / chrF++ per domain, pooled-adapter ablation
FR7	Answer, ask for clarification, or abstain when information is missing or the question is out of scope	clarification/ abstention controller	controller test set, end-to-end behaviour table
FR8	Never give emergency, dosage or prescription advice; always refer to a registered doctor	rule-based medical safety layer	safety test set, referral rate
FR9	Respond in Standard Bangla with a transparent trace (variety, normalized question, domain, decision, evidence)	pipeline output	chatbot demonstration
FR10	Produce every table, figure and statistical test of the thesis reproducibly, reported separately for each variety	staged notebooks + manifests	regenerated result files

3.1.2 Non-Functional Requirements

- **Fairness across varieties:** Every result is reported separately for Standard Bangla, Chittagonian and Sylheti, and the Standard-vs-dialect gap is a primary metric.
- **No harm to Standard input:** Dialect handling must not reduce quality for Standard Bangla questions (tested by the NO_NORMALIZATION and ALWAYS_NORMALIZE ablations).
- **Efficiency:** One 4-bit base model serves every domain; each additional domain costs one adapter of 133 MB instead of another large language model.

End-to-end latency is about 0.57 s per question in batched evaluation on one GPU.

- **Modularity:** The routers, normalization expert, retrievers, knowledge bases and domain adapters can each be replaced or retrained without retraining the others, so that new dialects or domains can be added later; one domain expert trains in 55–113 minutes.
- **Reproducibility:** Fixed seeds (17; 42 and 101 for finalists), pinned software versions, configuration files, SHA-256 manifests of every data transfer, and model selection on validation data only.
- **Robustness of execution:** Every GPU job is resumable from checkpoints, so interruption of a cloud machine never loses finished work.
- **Privacy:** No user data are stored; the knowledge bases are public resources and patient questions never leave the system (no external API).

3.1.3 Technical and Resource Requirements

- **Software:** Python 3.12, PyTorch 2.8.0 (CUDA 12.8), Hugging Face transformers 4.57.6 (and 5.17.0 in a separate environment for Gemma-4), PEFT 0.21.0, bitsandbytes 0.50.2, accelerate 1.15.0, scikit-learn 1.6.1 and sacreBLEU 2.5.1.
- **Retrieval stack:** BM25 for sparse retrieval and BGE-M3 and multilingual-E5 embeddings for dense retrieval [14], [62], [78], with BanglaBERT as the transformer baseline for the routers [40].
- **CPU compute:** Google Colab CPU runtimes (browser or VS Code) for data preparation, routers, sparse retrieval, analysis and report generation.
- **GPU compute:** One NVIDIA A40 (48 GB, 44.4 GB usable) rented on RunPod for model comparison, expert training, dense retrieval and full-system evaluation. All language-model training uses 4-bit quantization and LoRA adapters so that it fits in a single GPU [50].
- **Storage and transfer:** Google Drive for data and results, a private Hugging Face dataset repository for moving stage outputs between Colab and RunPod, and an 80 GB volume on the GPU machine.
- **Human resources:** Native speakers of Chittagonian and Sylheti to verify dialect questions and a physician to review medical answers. Review sheets for both were prepared, but these reviews could not be completed within the project time (Section 5.11).

3.1.4 Final Design Specification

Table 3.2 summarises the configuration that was finally selected.

Table 3.2: Final specification of the selected design.

Component	**Selected configuration**
Base language model	Gemma-4-E4B-it [102], loaded once in 4-bit NF4 with double quantization
Experts (Mixture-of-Adapters)	LoRA adapters [43] (r = 16, α = 32, dropout 0.05, all attention and MLP projections; 34.9 M trainable parameters, 133 MB each): normalization (2 epochs, selected on validation data), education, general, medical; hard top-1 routing
Dialect router	char (1–5) + word (1–2) TF-IDF, calibrated linear SVM; test macro-F1 0.946
Domain router	same features and classifier over education / general / medical / other; test macro-F1 0.908
Controller	gradient-boosted trees on router probabilities, retrieval scores and surface features; decisions ANSWER / CLARIFY / ABSTAIN
Retrieval	education and general: BM25; medical: dense BGE-M3 [62]; 3 passages per question
Knowledge bases	education 163,591 chunks, general 7,948, medical 28,134
Safety layer	rules for emergencies (fixed reply pointing to the national emergency number 999), dose/prescription refusal, referral note on every medical answer
Output	Standard Bangla answer + trace (variety, normalized question, domain, decision, evidence)

3.2 Societal Impact

Chittagonian and Sylheti are spoken by millions of people in Bangladesh and the diaspora, yet question answering systems and language models are developed and evaluated predominantly on Standard Bangla. Our results quantify the consequence: without normalization the same questions score 33% (Chittagonian) and 43% (Sylheti) lower than in Standard Bangla (Section 5.6). The expected positive impacts are:

- **Access to health information:** People can describe symptoms in their own variety and receive evidence-grounded general health information, with emergencies redirected to 999 and a doctor referral on every medical answer. This matters in a country where only about 11% of doctors and nurses work in rural areas, home to about two thirds of the population [45].
- **Education support:** Students can ask textbook questions (national curriculum) in their home variety and receive answers grounded in the same textbooks they study [99].

- **Digital inclusion:** Speakers who write in romanized Bangla or in a regional variety are served by the same system, instead of being expected to switch to the standard written form.
- **Research infrastructure:** The evaluation protocol (paired questions in three varieties, per-variety reporting, human-written dialect test questions) makes the dialect gap measurable and can be reused for other Bangla varieties, together with the trained routers, normalization adapter and evaluation scripts.

The main societal risks are over-reliance on automatic health answers and wrong answers in dialects the system does not know well. They are mitigated by the informational-only design, abstention when evidence is missing, the referral note, and transparent reporting of failure cases. The system is a research prototype and is not intended for clinical use without further validation by physicians, regulatory review and ongoing monitoring.

3.3 Ethical Issues

- **Medical safety:** The medical branch is informational only and the system does not present itself as a doctor. Emergency descriptions bypass the language model and return a fixed message to call 999; dose or prescription requests are refused; every medical answer ends with a referral to a registered doctor. Advice that could delay urgent care is treated as the most serious type of error. No language model is used to judge safety; a physician review sheet was prepared for human validation.
- **Data licensing and attribution:** Only public research datasets were used and each is cited. Several (e.g. BanglaRQA [41], released under CC BY-NC-SA 4.0, and BanglaCHQ-Summ [55]) carry non-commercial licences, so the trained system is for research use. The dialect health dataset BanglaCHQ-Prantik [86] was not used without the authors' permission.
- **Privacy:** No personal data were collected. Knowledge-base texts come from published resources; the demonstration stores no conversations. Closed commercial models were deliberately not used, so that health questions never leave the system.
- **Fairness and representativeness:** Performance is reported per variety, and the remaining dialect gap is stated explicitly rather than averaged away. Dialect data are imbalanced across varieties, and some sources, such as NCTB-QA, contain model-generated questions [99]; these limitations are stated with the results.
- **Respect for dialects:** Normalizing a dialect into Standard Bangla is a technical step to reuse existing resources. It does not imply that regional varieties are incorrect or inferior; answering in the user's own dialect is listed as future work.

- **Transparency and trust:** Every answer is returned with a trace (detected variety, normalized question, domain, decision and evidence), and the system abstains or asks for clarification when it is not sure.
- **Honest reporting:** Negative and neutral findings are reported: the dialect-invariant encoder did not beat its base encoder, routed experts do not beat a single pooled adapter on accuracy, and the Chittagonian result remains below a published specialised model.

3.4 Standards

The work follows these standards and guidelines (alignment with their principles; no formal certification is claimed):

- **Text encoding:** All text is stored in UTF-8 and normalized to Unicode Normalization Form C (Unicode Standard Annex #15) with invisible characters removed, so that identical Bangla strings compare equal.
- **Evaluation reporting:** Corpus BLEU and chrF++ are computed with sacreBLEU [23]; confidence intervals and paired significance tests follow common NLP practice [9], [12]; multiple comparisons are corrected with the Holm procedure [2].
- **Reproducibility and data management:** Fixed train/validation/test splits (official splits preserved), fixed random seeds, pinned package versions, versioned configuration files, a leakage audit, and SHA-256 checksums for every transferred file.
- **Responsible AI for health:** The safety design follows widely accepted principles for AI in health information, including the World Health Organization's guidance [39]: the system gives general information only, defers diagnosis and prescribing to qualified professionals, and escalates emergencies.

3.5 Project Management Plan

The project is organized into phases aligned with the thesis timeline. The plan prioritizes early data readiness and a working baseline pipeline, followed by component improvement and evaluation.

- **Phase 1: Literature review and planning**

  Finalize the scope, target dialects, domains and evaluation metrics, and confirm the availability and licenses of the datasets.

- **Phase 2: Data collection and preprocessing**

  Collect the dialect, QA and healthcare datasets, clean and normalize them, build the unified data schema, and construct the three knowledge bases (education, general knowledge and medical).

- **Phase 3: Router and normalization expert**

  Train and evaluate the variety and domain router, and train the dialect normalization adapter and compare it with zero-shot and few-shot prompting baselines.

- **Phase 4: Domain RAG experts and base-model comparison**

  Build and select the retriever of each domain (BM25 or dense), train the healthcare, education and general-knowledge adapters, and compare the candidate base models.

- **Phase 5: Dialectal test set, evaluation and ablations**

  Build the paired dialect test set, run the end-to-end and ablation experiments, and evaluate the system on dialect questions written by native speakers.

- **Phase 6: Writing and finalization**

  Consolidate the results, prepare the final figures and tables, and complete thesis formatting and proofreading.

Current status. All six phases have been completed. The data stage cleaned and split more than twenty public sources into dialect–Standard pairs, QA items and three knowledge bases (education, general knowledge and medical), and an automatic leakage audit found no overlap between training, knowledge-base and test data. The variety and domain routers and the dialect normalization expert were trained and evaluated (dialect-router macro-F1 0.946; normalization chrF++ 76.3 for Chittagonian and 74.0 for Sylheti). Seven open base models were compared under one QLoRA budget, Gemma-4-E4B-it was selected, and the domain experts and retrievers were trained and chosen on validation data. The complete system was evaluated under nine conditions on 2,225 paired test questions and on 481 dialect questions written by native speakers, and the thesis is in its final writing stage.

3.6 Risk Management

The project faced both technical and practical risks. The mitigations were planned in advance, and Table 3.3 also records what actually happened.

Table 3.3: Risks, mitigations and what actually happened.

Risk	Mitigation	Outcome
Cloud GPU stopped or lost	resumable jobs with atomic checkpoints; results pushed to a private repository after every stage	pods were stopped and replaced several times without losing work
Newest model not loadable by the stable software stack	separate environment with transformers 5 for Gemma-4	Gemma-4 trained and evaluated normally

Risk	Mitigation	Outcome
Out-of-memory or silently truncated training data	4096-token limit after a tokenizer-fertility audit; per-model count of dropped examples	0 examples dropped in every expert
Test data leaking into training or knowledge bases	group-wise splits and an automatic leakage audit	0 leaks in all six audits
Moving to a new cloud account	migration notebook comparing rows read per source before and after	identical data on the new account
Native-speaker dialect test data unavailable	human-written question sentences of held-out corpus test splits	481 human-written test questions evaluated
Harmful medical output	rule-based safety layer, referral note, abstention	emergencies handled; remaining gaps reported (Section 5.9)
Budget overrun	cost-aware schedule, CPU for all non-GPU work, deleting storage after use	total GPU cost about US$20
Too little Chittagonian data, especially in the medical domain	combine all available dialect sources; one normalization adapter for both dialects with the variety named in the prompt	16,030 Chittagonian training pairs; medical Chittagonian remains the scarcest
Normalization changes the meaning of a question	confirmation rule for uncertain normalizations; comparison with base-model and prompting baselines	chrF++ 76.3 (Chittagonian) and 74.0 (Sylheti); the confirmation rule never fired, and some dialect words were kept literally
Router confuses varieties that share features	calibrated router; every dialect label triggers normalization	macro-F1 0.946; 97.4% / 94.3% of human-written Chittagonian / Sylheti recognised, but only 71.5% of synthetic Chittagonian
Routed design does not outperform a single pipeline	pooled adapter with the same number of training steps as a control	equal accuracy (± 0.005); the value of routing is modularity

3.7 Economic Analysis

Development used free CPU notebooks and about 40 rented A40 GPU-hours at US$0.49 per hour, i.e. about US$20 of GPU time plus a few dollars of storage, well within a student budget. Serving is also inexpensive: in batched evaluation the complete pipeline answered 2,225 questions in 1,268 s on one A40 (about 0.57 s per question), i.e. roughly 6,300 questions per GPU-hour, or well under US$0.001 per question at the same price. Because all domains share one quantized base model, adding a domain requires training one adapter (55–113 minutes in our runs) rather than hosting another model, and an existing domain can be updated without retraining the others (55 minutes for the medical expert versus 317 minutes for a single pooled adapter covering all tasks). Open models also avoid per-token API fees and keep health questions local. Further economic benefits are:

- **Reuse instead of new data collection:** Normalizing dialectal questions lets the system reuse existing Standard Bangla textbooks, QA datasets and health content instead of building separate resources for every dialect.
- **Potential savings for users:** Reliable first-line health information and curriculum-aligned learning support in a user's own variety could reduce unnecessary travel, delayed care and reliance on paid private tutoring.

Chapter 4

Proposed Methodology

4.1 Introduction and Design Principles

A question such as "what are the signs of dengue?" can reach a Bangla assistant in Standard Bangla, in Chittagonian, in Sylheti, or typed in Latin letters. Existing Bangla QA resources and language models are Standard-only [41], [90], [99], and in our own measurement the same questions asked in dialect scored 33% (Chittagonian) and 43% (Sylheti) lower when the system did not normalize them. The methodology in this chapter is designed to close that gap without building a separate system for every dialect. It rests on four principles.

P1 – Route, normalize, then ground. Identify the language variety first, rewrite dialect and romanized questions into Standard Bangla, and only then choose a domain, retrieve evidence and answer. This lets every Standard Bangla resource (textbooks, Wikipedia passages, doctor answers) be reused unchanged.

P2 – One base model, many small experts. All language-model skills (normalization, education, general knowledge, medicine) are low-rank adapters on *one* 4-bit base model, selected by lightweight routers: a *Mixture-of-Adapters*. Adding or updating a domain means training one adapter.

P3 – Decide on DEV, report on TEST, per variety. Every choice (base model, retriever, thresholds, number of epochs) is made on validation data. Test data are used once for reporting, and every result is reported separately for Standard Bangla, Chittagonian and Sylheti.

P4 – Three data roles that never mix. *Training data* change weights, *knowledge* is only searched, and *evaluation data* are held out; an automatic audit checks six possible leakage paths.

Fig. 4.1 shows the complete system, and Table 4.1 lists the research questions and the experiments that answer them. Section 4.2 gives the background, Section 4.3 the overall workflow, Section 4.4 the data, Section 4.5 the base-model selection, Section 4.6 each component, Section 4.7 the evaluation protocol and Section 4.8 the implementation environment.

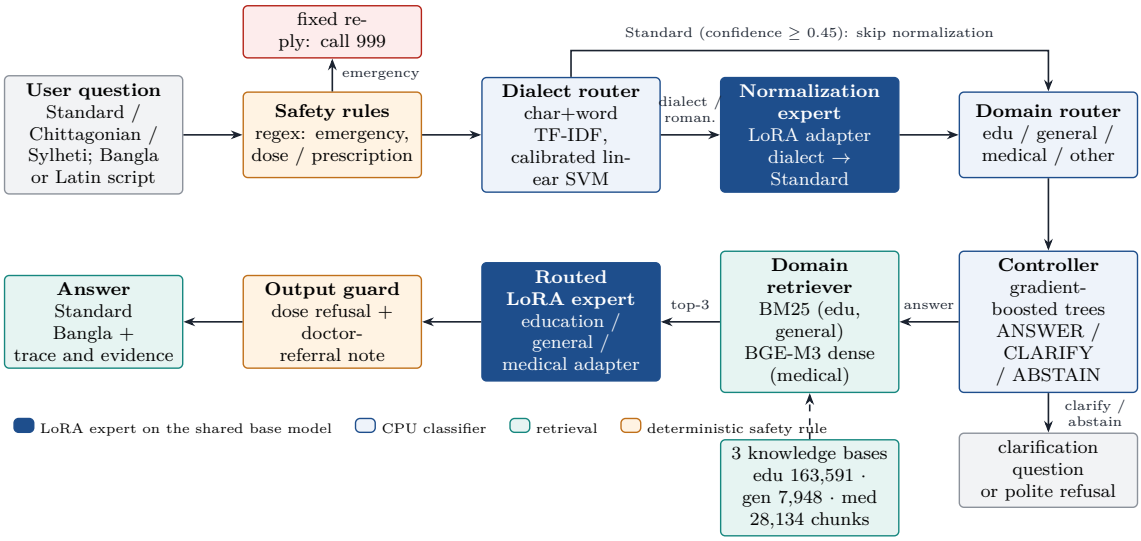


Figure 4.1: Inference pipeline. A question passes the safety rules, is routed by language variety, normalized if it is dialectal or romanized, routed by domain, checked by the controller, grounded in three passages from its domain's knowledge base, answered by that domain's LoRA expert and finally checked by the output guard. The two dark boxes are LoRA adapters on one shared 4-bit Gemma-4-E4B-it model and are the only steps that run the language model; everything else runs on a CPU.

Table 4.1: Research questions and the experiments (stages) that answer them.

	Research question	Experiment (stage)
RQ1	How accurately, and with how well-calibrated confidence, can a lightweight router identify the variety and the domain?	router comparison incl. BanglaBERT (S4); routing inside the system (S5); human-written questions (S6)
RQ2	How well can a fine-tuned model normalize dialect and romanized questions, and how much of the dialect gap does it close?	normalization benchmark, prompting baselines (S3, S5); `FULL` vs `NO_NORMALIZATION` (S5); S6; S8
RQ3	What do retrieval, fine-tuned experts, domain routing and the controller contribute; do routed experts beat one shared adapter?	ablations `NO_RAG`, `BASE_MODEL`, `NO_DOMAIN_ROUTING`, `NO_CONTROLLER`, `POOLED_ADAPTER` (S5)
RQ4	Which open base model gives the best quality–cost balance, and does Bangla pre-training help after fine-tuning?	7 models, identical QLoRA budget, 3 seeds for finalists (S2)

4.2 Background

4.2.1 Bangla Varieties and Dialect Normalization

Standard Bangla (*cholito bhasha*) is the variety of textbooks, news and administration. Regional varieties differ from it in pronunciation, vocabulary, verb endings and sometimes word order; Chittagonian (south-east) and Sylheti (north-east) are

Table 4.2: Dialect words and their Standard Bangla forms (transliterated; from the TRAIN split of the word-level glossary, Section 4.7.7).

Variety	Dialect form	Standard form	Meaning
Chittagonian	*ãi*	*ami*	I
Chittagonian	*tun*	*theke*	from
Chittagonian	*baiddhe*	*bedheche*	has tied
Sylheti	*kita*	*ki*	what
Sylheti	*koilam*	*bollam*	(I) said
Sylheti	*jaimu*	*jabo*	(I) will go
Sylheti	*pushkuni*	*pukur*	pond

among the most divergent [51], [103]. Because there is no agreed dialect spelling, writers spell words as they are pronounced, and many type Bangla in Latin letters. *Dialect normalization* rewrites a dialect sentence into Standard Bangla with the same meaning, so it is a form of machine translation [51]. Table 4.2 shows word pairs from the training split of our dialect glossary: most everyday words change, which is why a system trained on Standard Bangla misreads dialect input.

4.2.2 Large Language Models and Instruction Tuning

Modern language models are decoder-only Transformers [19] pre-trained to predict the next token and then instruction-tuned, i.e. fine-tuned on (instruction, response) pairs so that they follow a task description [46]. We phrase every task as a chat instruction in Bangla: a system message states the task, the user message carries the input, and the target is the desired output. A model's *tokenizer fertility* (tokens per word) decides how long a Bangla sentence becomes and therefore the training cost and the context that fits.

4.2.3 Low-Rank Adaptation (LoRA)

Updating every weight of a billion-parameter model is expensive. LoRA [43] freezes a pretrained weight matrix $W_0 \in \mathbb{R}^{d\times k}$ and learns a low-rank update:

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} BAx, \tag{4.1}$$

with $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$ and $r \ll \min(d, k)$. Only A and B are trained ($r(d + k)$ instead of dk values per matrix; B starts at zero so training starts from the original model). Many adapters can be attached to one frozen base and switched per request (Fig. 4.2). Our adapters use $r = 16$, $\alpha = 32$ in every attention and MLP projection: 34.9 M trainable parameters (133 MB) each.

4.2.4 QLoRA: Fine-Tuning a Quantized Model

QLoRA [50] stores the frozen base in 4-bit NormalFloat (NF4), a data type whose 16 levels are quantiles of a normal distribution, so it fits normally distributed weights well; the quantization constants are themselves quantized (*double quantization*). Gradients flow through the de-quantized weights into the LoRA matrices, which are

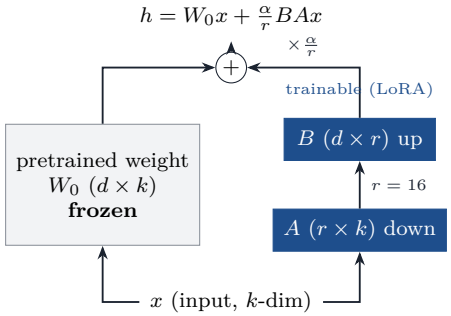


Figure 4.2: LoRA: the pretrained weight stays frozen; only the low-rank matrices A and B are trained. Swapping (A, B) swaps the skill while the base model stays in memory once.

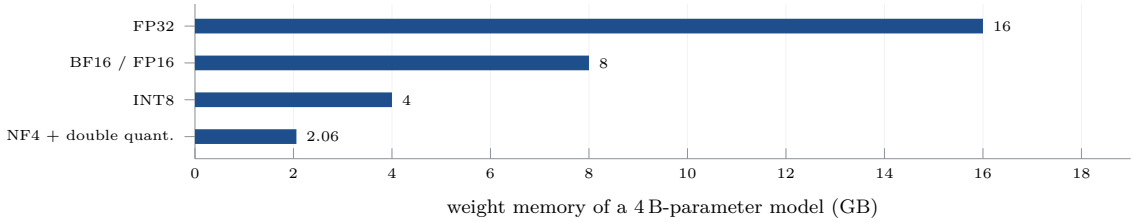


Figure 4.3: Weight memory per numeric format for 4×10^9 parameters (bytes per parameter: FP32 4, BF16 2, INT8 1, NF4 with double quantization ≈4.127 bits [50]). Activations, the LoRA weights and the optimizer come on top.

kept in 16-bit precision. Fig. 4.3 shows the weight memory of a 4-billion-parameter model: NF4 needs about a quarter of the 16-bit memory, which is what lets one rented 48 GB GPU train and serve every expert.

4.2.5 Retrieval-Augmented Generation

A language model that answers from memory can produce fluent but unsupported answers [53]. Retrieval-augmented generation (RAG) first retrieves passages relevant to the question and conditions the generator on them [33] (Fig. 4.4). Two kinds of retrievers are used.

Sparse retrieval (BM25) scores a document d for a query q by weighted term overlap [14]:

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t)\, \frac{f_{t,d}\,(k_1+1)}{f_{t,d} + k_1\big(1 - b + b\,\frac{|d|}{\overline{|d|}}\big)}, \tag{4.2}$$

with $\mathrm{IDF}(t) = \ln\big(1 + \frac{N - n_t + 0.5}{n_t + 0.5}\big)$, term frequency $f_{t,d}$, N documents of which n_t contain t, $k_1 = 1.2$ and $b = 0.75$. We add light Bangla suffix stripping.

Dense retrieval encodes query and passage into vectors and ranks by cosine similarity $\cos(\mathbf{e}_q, \mathbf{e}_d)$ [31]; BGE-M3 [62] and multilingual-E5 [78] support Bangla. **Hybrid** retrieval fuses ranked lists with reciprocal rank fusion [13]:

$$\mathrm{RRF}(d) = \sum_{m} \frac{1}{k + \mathrm{rank}_m(d)}, \qquad k = 60. \tag{4.3}$$

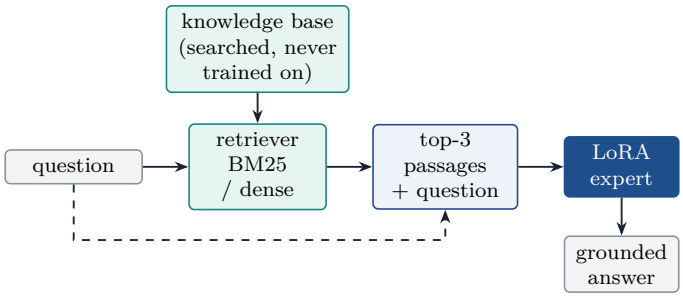


Figure 4.4: Retrieval-augmented generation: the expert answers from retrieved evidence, or writes a fixed "not found" sentence when the evidence does not contain the answer.

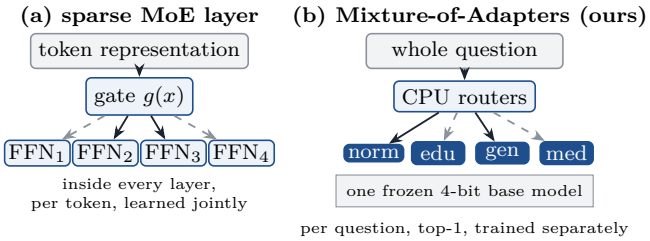


Figure 4.5: Token-level sparse MoE vs. the system-level Mixture-of-Adapters used here: one adapter for rewriting, then the adapter of the predicted domain for answering.

4.2.6 Mixture of Experts and the Mixture-of-Adapters

A mixture of experts combines specialised models through a gating function [3]; in modern LLMs, sparse MoE layers route each *token* to a few expert sub-networks inside one model [18], [42]. We apply the idea at the *system* level (Fig. 4.5): CPU routers send each *question* to exactly one LoRA adapter on a shared base model (hard top-1 routing), closer to adapter-based transfer [25] and to routed retrieval [67], [85] than to an MoE layer. The experts stay small, interpretable and replaceable.

4.2.7 Calibrated Classifiers

The routers must say *how sure* they are, because low confidence changes what the system does. A linear support vector machine [5] outputs a score $f(x)$, which sigmoid (Platt) calibration [6] turns into a probability $P(y{=}1 \mid x) = 1/(1{+}\exp(Af(x){+}B))$, with A, B fitted on held-out folds. Calibration quality is measured by the expected calibration error [16]:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{n} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|, \tag{4.4}$$

where the predictions are grouped into M confidence bins B_m; a well-calibrated router is right 90% of the time when it says 0.9.

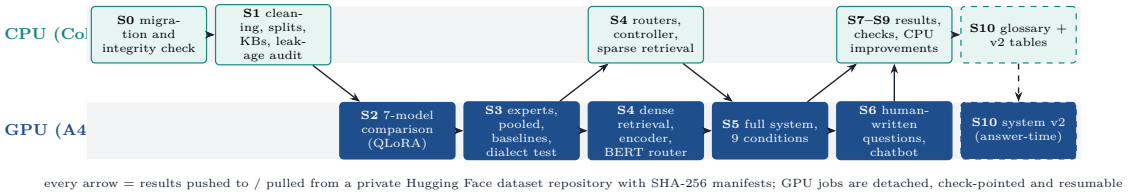


Figure 4.6: Development workflow. Teal stages run on a CPU, dark stages on one rented A40 GPU; dashed stages are the optional DEV-selected improvements (Section 4.7.7). About 40 GPU-hours (US$20) were used up to S6.

4.2.8 Selective Answering

A realistic system must also recognise questions it should not answer. SQuAD 2.0 showed that models trained only on answerable questions answer unanswerable ones confidently [24], and selective QA trades coverage for accuracy by abstaining when unsure [30]. Our design separates three cases: the question is out of scope or vague (the controller asks or refuses), the evidence lacks the answer (the expert writes a fixed "not found" sentence, learned from real unanswerable questions), or it is an emergency (a fixed safety reply).

4.3 Overall Workflow

The methodology follows *compare* → *adapt* → *route* → *evaluate*. We compare open base models under an identical fine-tuning budget, adapt the winner into LoRA experts, build the routers, retrievers, controller and safety layer around them, and evaluate the whole system and its ablations on the same questions in three varieties. The work runs as staged notebooks (Fig. 4.6): stages that only need a CPU run on free Colab runtimes, and stages that train or run the language model run as resumable background jobs on one rented NVIDIA A40 (48 GB). Stage outputs move between the two through a private Hugging Face dataset repository with SHA-256 manifests. Algorithm 1 summarises inference.

4.4 Data

4.4.1 Three Data Roles

Fig. 4.7 shows how the data are separated. *Training data* change model weights: dialect–Standard pairs, the training splits of the QA datasets, and the router and controller data. *Knowledge* is only searched at answer time: textbook chunks, Wikipedia passages, doctor answers from the training split, and drug and disease descriptions. *Evaluation data* are held out: DEV for every choice, TEST once for reporting.

Algorithm 1 Answering one question

Require: question q; routers $R_{\text{var}}, R_{\text{dom}}$; controller C; KBs $\mathcal{K}_d$; experts E_{norm}, E_d

if Emergency(q) **then return** fixed emergency reply (call 999)
$(v, p_v) \leftarrow R_{\text{var}}(q)$ ▷ Standard / Chittagonian / Sylheti
if ($v \neq$ Standard **and** $p_v \geq 0.45$) **or** q is romanized **then**
    $\hat{q} \leftarrow E_{\text{norm}}(q, v)$ ▷ LoRA normalization
else
    $\hat{q} \leftarrow q$
if Emergency($\hat{q}$) **then return** fixed emergency reply
$(d, p_d) \leftarrow R_{\text{dom}}(\hat{q})$ ▷ edu / general / medical / other
$a \leftarrow C$(router probabilities, BM25 signals, text features)
if $a =$ CLARIFY **then return** clarification question
if $a =$ ABSTAIN **or** $d =$ other **then return** polite refusal
$P \leftarrow$ top-3 passages of $\hat{q}$ in $\mathcal{K}_d$ ▷ BM25 or dense
$y \leftarrow E_d(P, \hat{q})$ ▷ expert of domain d; may say "not found"
if $d =$ medical **then**
    $y \leftarrow$ [dose refusal if requested] $+\, y +$ doctor-referral note
return y with trace $(v, p_v, \hat{q}, d, a, P)$

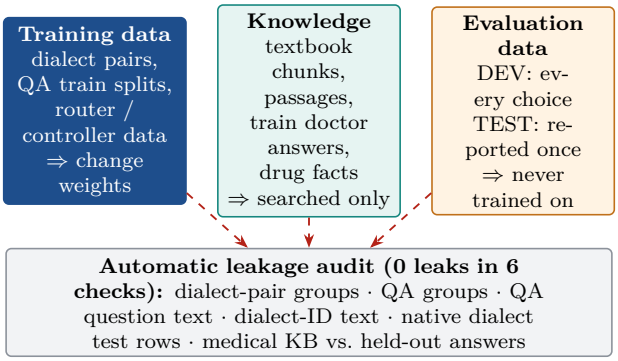


Figure 4.7: The three data roles and the audit that keeps them apart. Test passages of NCTB-QA and BanglaRQA are in the KB (open-book QA needs them), but test questions and medical test answers never are.

4.4.2 Sources

Table 4.3 lists every source and its role. Dialect data come from Vashantor [51], Kothon [100] and the ChatgaiyyaBench corpus [96], which contains the ChatgaiyyaAlap [82], ONUBAD [93] and BD-dialect sentences. The Mendeley release of ChatgaiyyaAlap (`Dataset_Chittagong_2.0.csv`) holds the same sentences as the ChatgaiyyaAlap part of ChatgaiyyaBench, so all of its rows were removed as duplicates. Fig. 4.8 shows how many dialect–Standard pairs each source contributes after cleaning.

4.4.3 Exploratory Data Analysis

The dialects differ strongly from Standard Bangla at the surface (Fig. 4.9 and Fig. 4.10). A Chittagonian sentence keeps only 29.6% of the words and 43.0% of the character trigrams of its Standard translation; for Sylheti the figures are 36.5% and 49.4%. Measured as Jaccard similarity (shared trigrams divided by *all* distinct

Table 4.3: Data sources, rows used, and role ("rows used" are counted before cross-source de-duplication; Fig. 4.8 shows the unique pairs).

Source	Rows used	Role
Vashantor [51] (Chittagong, Sylhet; Barishal, Mymensingh, Noakhali)	12,498	dialect pairs incl. romanized; the three other regions are held out as *unseen* dialects (test only)
Kothon [100] (Chittagonian, Sylheti)	17,328	dialect pairs (native annotated)
ChatgaiyyaBench corpus [96] (ChatgaiyyaAlap [82], ONUBAD [93], BD-dialect sentences; translation benchmark; romanized)	21,614	Chittagonian and Sylheti pairs, romanized pairs; its translation benchmark is test only
ChatgaiyyaAlap, Mendeley release (`Dataset_Chittagong_2.0.csv`)	0 new	identical to the ChatgaiyyaAlap part of ChatgaiyyaBench: removed as duplicates
Local Language Dataset (regional dialects)	1,007	unverified pairs: *training only*, never evaluated
NCTB-QA [99]	87,805	education QA with unanswerable questions (official split) + KB passages
BanglaRQA [41]	14,889	general-knowledge QA (official split) + KB passages
Bangla national-curriculum textbooks	131,361	education KB chunks (retrieval only)
Doctor QA (Bangla)	4,417	medical QA (80/10/10 grouped split) + KB (training answers only)
Bangla-MedER [97]; Medicine dataset of Bangladesh; healthcare-chatbot intents	24,725	medical KB (retrieval only)
Bangla-HealthNER [56]; specialist classification; BanglaCHQ-Summ questions [55]	25,782	domain-router data
Out-of-scope news and blog sentences	8,006	domain router "other" class
BEnQA [75]; Bangla-MMLU sample; BanglaMedQA [94]	8,075	multiple-choice probes (evaluation only)
ChatgaiyyaBench QA cloze	590	native Chittagonian comprehension probe (evaluation only)

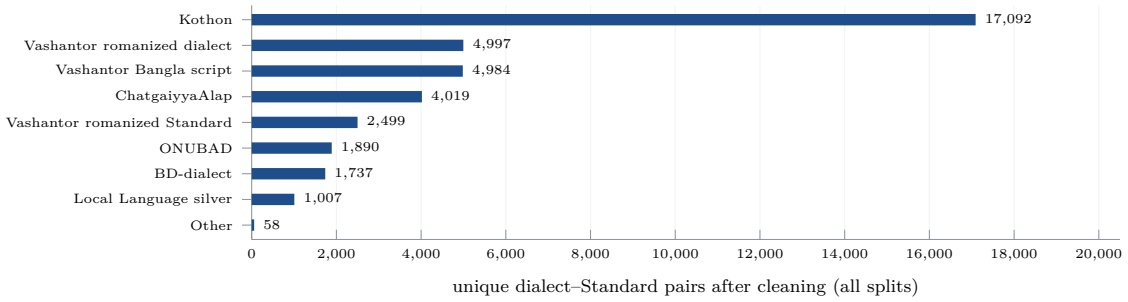


Figure 4.8: Unique dialect–Standard pairs per source after cleaning and cross-source de-duplication (38,283 in total; 16,905 Chittagonian and 13,855 Sylheti in Bangla script, 7,523 romanized).

trigrams of both sentences), the overlap is 0.296 and 0.348, so Chittagonian is the harder variety. Sentences are short (about 6–7 words), so changing two or three words already removes half of the trigrams. Romanized questions share no words with Bangla-script text at all, because they use another alphabet, which is why romanized input is always normalized. The QA data differ by domain: education questions average 9.8 words with 10.4-word answers, general questions 8.2 and 5.2 words, and medical questions 18.6 words with 50.1-word doctor answers; 43% of education and 24% of general questions are unanswerable, which gives real training signal for abstention.

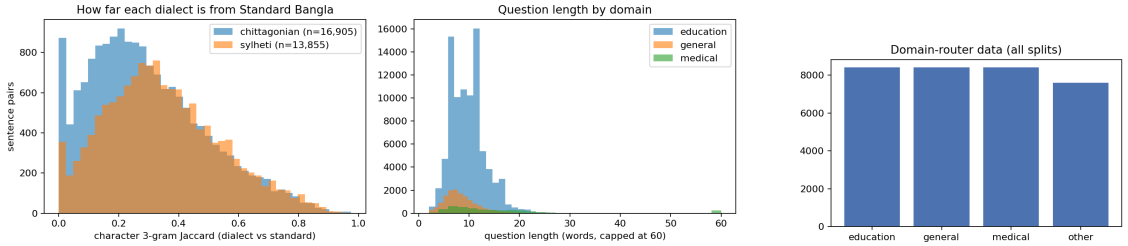


Figure 4.9: Left: character 3-gram Jaccard between each dialect sentence and its Standard translation (a peak near 0 means almost every trigram changed), and question length per domain. Right: the balanced domain-router data.

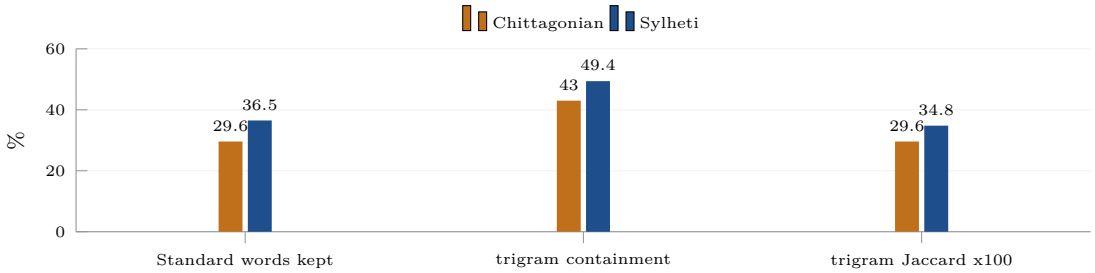


Figure 4.10: How much of the Standard sentence survives in the dialect sentence (Bangla-script pairs, all splits).

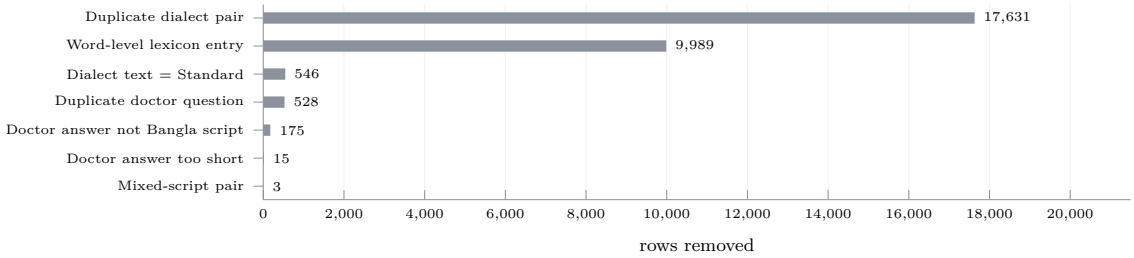


Figure 4.11: Cleaning log. Duplicates dominate: ChatgaiyyaBench re-distributes Vashantor and ChatgaiyyaAlap sentences. Doctor QA shrinks from 5,135 to 4,417 pairs.

4.4.4 Cleaning

All text is Unicode-NFC normalized; invisible characters (zero-width space, soft hyphen, byte-order mark) are removed while the joiners Bangla needs are kept. Fig. 4.11 lists the removals. Word-level dictionary entries are removed from the *training* pairs because the normalization expert rewrites whole questions, and 10k one-word examples would teach context-free word swaps and very short outputs; they were reused later, differently, as a glossary of 2,356 Chittagonian and 974 Sylheti entries (TRAIN split only), which was tested in system v2 (Section 4.7.7). Pairs whose dialect text equals the Standard text carry no dialect signal and would give the dialect router the same sentence with two labels.

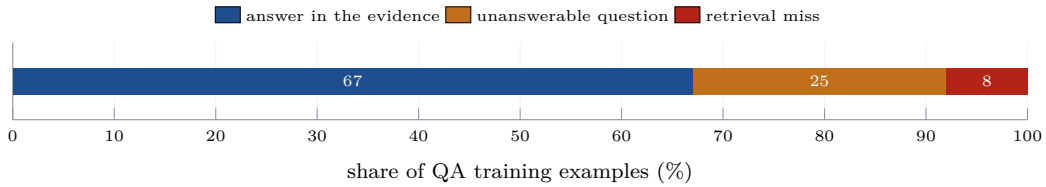


Figure 4.12: Target mix of the QA experts' training data: a third of the targets teach the expert to say "not found" instead of guessing (RAFT-style training [81]).

Table 4.4: Leak-free split sizes.

Task	Subset	Train	DEV	TEST
Dialect pairs	Chittagonian	16,030	1,411	1,968
	Sylheti	13,119	1,295	1,961
	Standard (romanized)	1,874	250	375
Dialect ID	Chittagonian	12,000	1,403	1,957
	Standard	12,000	2,000	2,000
	Sylheti	12,000	1,275	1,947
Domain router	4 classes	6,000 each	1,200	1,200
QA	Education	70,303	8,119	8,489
	General	11,912	1,441	1,446
	Medical	3,513	433	471
MCQ (eval only)	3 benchmarks	–	2,457	5,618
Controller	3 decisions	–	1,127	1,857

4.4.5 Transformation

Every task becomes a chat instruction (system message, user message, target), written in Bangla so that the model answers in Bangla. Normalization prompts name the variety ("variety: Chittagonian; sentence: ..."); QA prompts contain three retrieved passages followed by the question. To teach grounded behaviour, the QA training targets follow the mix in Fig. 4.12: 25% of examples are real unanswerable questions and 8% are *retrieval misses* (the gold passage removed), both with a fixed "the answer was not found in the given information" target. Knowledge-base passages are split into overlapping windows of 110 words (stride 90) for NCTB-QA and BanglaRQA passages and 128 words for textbook rows (about 93–95 words on average); medical documents are stored as question–answer pairs or entity descriptions (about 27 words).

4.4.6 Integration, Splits and Reduction

The dialect sources are merged into one pair table with provenance and split by *group*: all versions of the same Standard sentence stay in one split (80/8/12), so no Standard sentence is seen in training and testing. Official splits (Vashantor, NCTB-QA, BanglaRQA, ChatgaiyyaBench benchmarks) are preserved, and a question text may never appear in two splits. Training sets are capped to fit the GPU budget: 24,000 normalization pairs, 11,943 education, 7,994 general and all 3,513 medical examples. Unverified silver data are used for training only. Table 4.4 gives the resulting splits and Table 4.5 the knowledge bases.

Table 4.5: Knowledge bases (searched, never trained on).

KB	Chunks	Documents	Avg. words
Education (NCTB-QA passages + textbooks)	163,591	96,907	93.2
General (BanglaRQA passages)	7,948	2,996	94.9
Medical (doctor answers, MedER, drugs, intents)	28,134	28,134	26.9
Medical evaluation pool (retrieval benchmark only)	904	904	48.7

Table 4.6: Candidate base models.

Model	Params (B)	Bangla-adapted
TigerLLM-1B-it [90]	1.0	yes
Gemma-3-1B-it [83]	1.0	no
TituLM-Llama-3.2-3B [88]	3.2	yes (base model)
Llama-3.2-3B-Instruct [72]	3.2	no
Tiny-Aya-Fire [105]	3.35	no
Qwen3-4B-Instruct-2507 [95]	4.0	no
Gemma-4-E4B-it [102]	4.0	no

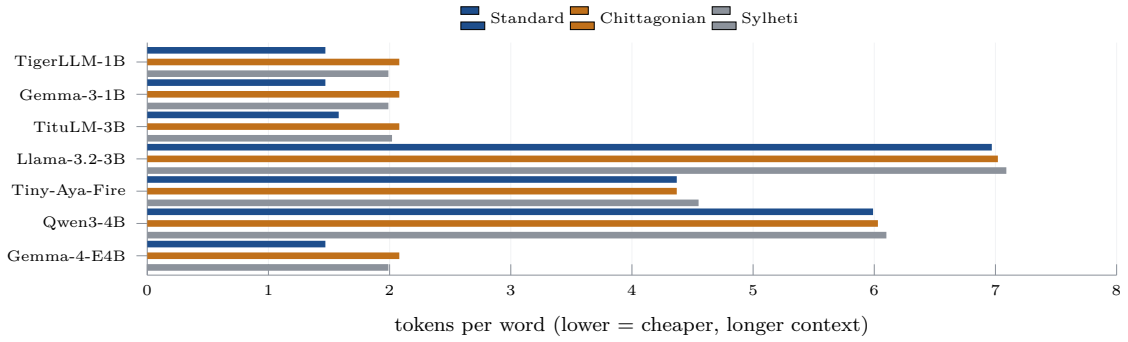


Figure 4.13: Tokenizer fertility on Standard, Chittagonian and Sylheti text. Dialect text costs 30–40% more tokens than Standard text for the Gemma tokenizers.

4.5 Base-Model Selection

Seven open models were compared (Table 4.6): six instruction-tuned models and one Bangla base model. They include two *Bangla-adapted vs. original* pairs (TigerLLM-1B [90] vs. Gemma-3-1B [83]; TituLM-3B [88] vs. Llama-3.2-3B [72]) to test whether Bangla continued pre-training still helps after task fine-tuning. The released TigerLLM-1B checkpoint has the architecture and tokenizer of Gemma-3-1B (identical fertility in Fig. 4.13), so Gemma-3-1B is its original model. Tokenizer fertility was measured first (Fig. 4.13): models that need 6–7 tokens per Bangla word make every example 4–5 times longer, which raises training time and forced a 4,096-token limit so that no long medical example is silently dropped.

Identical budget. Every candidate receives the same QLoRA budget: 400 optimizer steps on the same 6,800 mixed-task examples, the same LoRA configuration, learning-rate schedule and seed. All candidates are evaluated zero-shot and after fine-tuning on a shared benchmark, and the choice uses a DEV composite

$$S = \frac{1}{5}\left(\frac{\mathrm{chrF}_{\mathrm{ctg}}}{100} + \frac{\mathrm{chrF}_{\mathrm{syl}}}{100} + F1_{\mathrm{edu}} + F1_{\mathrm{gen}} + \frac{\mathrm{chrF}_{\mathrm{med}}}{100}\right), \tag{4.5}$$

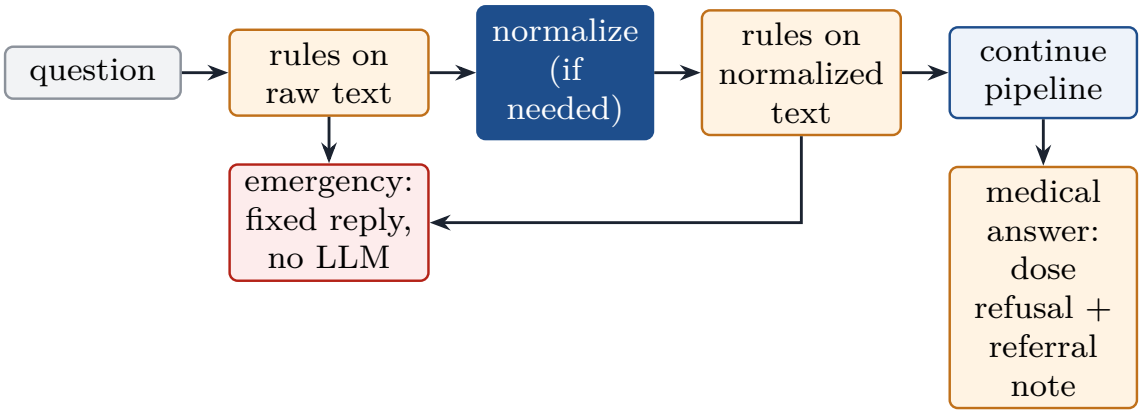


Figure 4.14: Deterministic safety: emergency rules before and after normalization, and the output guard on every medical answer.

where the first two terms are normalization quality and the rest are QA quality. The two finalists are re-trained with three seeds and compared with paired randomization tests; the winner (Gemma-4-E4B-it) hosts every expert.

4.6 System Components

4.6.1 Safety Layer

Regular expressions written for Standard Bangla, Chittagonian, Sylheti and romanized Bangla detect emergencies (e.g. chest pain with breathlessness, unconsciousness, poisoning, stroke signs, suicidal statements) and dose or prescription requests. They run twice, on the raw and on the normalized question (Fig. 4.14). Emergencies bypass the language model and return a fixed message asking the user to call the national emergency number 999 or go to the nearest hospital. No language model decides safety.

4.6.2 Dialect Router and Normalization Trigger

The dialect router classifies a question as Standard, Chittagonian or Sylheti from the union of character n-grams (1–5, word-bounded, robust to spelling variation) and word n-grams (1–2), weighted by TF-IDF, with a class-balanced linear SVM whose scores are calibrated with 3-fold sigmoid calibration; the regularisation constant C is tuned on DEV macro-F1. Logistic regression, Complement Naive Bayes and single-feature variants are the alternatives, and a fine-tuned BanglaBERT [40] tests whether a transformer router is worth its cost. The normalization trigger is

$$\text{normalize}(q) = \big[v \neq \text{Std} \wedge p_v \geq 0.45\big] \vee \big[q \text{ is romanized}\big], \tag{4.6}$$

so a Standard question is never rewritten unless the router is at least moderately sure it is dialectal.

4.6.3 Normalization Expert

The normalization expert is a LoRA adapter trained on 24,000 examples drawn from the training pairs in both directions: dialect or romanized → Standard (Chittagonian, Sylheti, and romanized input including romanized Standard Bangla) and Standard → dialect. The variety is named in the prompt; 15% of dialect prompts carry a generic "regional Bangla" label instead, so the expert also works when the router is unsure. It was trained for one epoch (1,500 steps) and, as a DEV-selected improvement, re-trained for two epochs; the two-epoch expert was adopted because it was better on DEV (78.67 vs. 77.24 chrF++). The reverse direction is used only to build the paired test set (Section 4.7.3). When the expert's mean token log-probability for a rewrite is below -0.9, the system shows its interpretation and asks the user to confirm it.

Baselines. Zero-shot prompting of the base model, and 5-shot prompting whose examples are the five most similar training pairs (character 2–4-gram TF-IDF, same variety and script), a stronger baseline than fixed examples; the BanglaCHQ-Prantik metric set [86] is reported next to chrF++. A LoRA expert on a 16-bit base with identical data, steps and seed answers the LoRA-vs-QLoRA question.

4.6.4 Domain Router

The domain router uses the same features and classifier over four classes (education, general, medical, other), trained on 6,000 balanced questions per class. It runs on the *normalized* question, so dialect words do not confuse it.

4.6.5 Answer / Clarify / Abstain Controller

The controller decides whether to answer, ask for clarification or abstain from 20 features (Fig. 4.15). Logistic regression and gradient-boosted trees [7] are compared with 5-fold cross-validation on DEV (macro-F1). Its data combine clear in-scope questions (ANSWER), out-of-scope sentences (ABSTAIN) and constructed under-specified questions such as a symptom without context (CLARIFY). Abstention because the *evidence* lacks the answer is not the controller's job: the experts learn it from real unanswerable questions.

4.6.6 Retrieval per Domain

Each domain has its own knowledge base. Four retriever families are compared on DEV (Standard questions, document-level nDCG@10): BM25, character TF-IDF, dense retrieval with BGE-M3 or multilingual-E5-base, and hybrids fused by RRF (Eq. 4.3). The retriever is chosen *per domain*; a dense method is taken only if it beats the best sparse method by at least 0.01 nDCG@10, because dense search costs GPU time. The result is BM25 for education and general knowledge and dense BGE-M3 for medicine. The expert receives the top $k = 3$ passages (at most 600 characters each, 1,800 in total).

Dialect-invariant encoder. As a novel variant, BGE-M3 is fine-tuned so that a

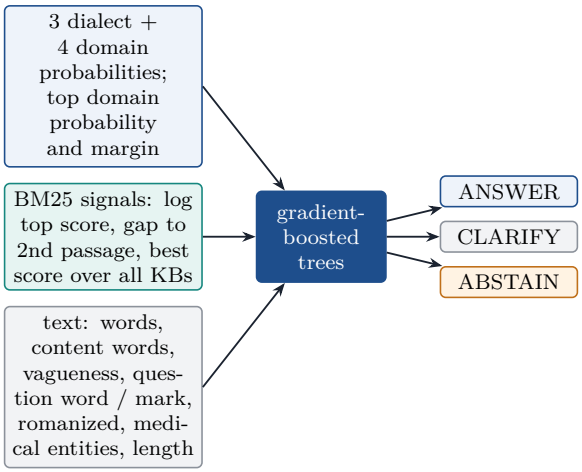


Figure 4.15: The controller's 20 features and its three decisions.

Table 4.7: Adapters trained on Gemma-4-E4B-it (QLoRA unless stated).

Adapter	Examples	Steps	Minutes	Peak GB
Normalization	24,000	1,500	72.9	19.9
Education	11,943	746	112.6	30.3
General	7,994	499	76.0	29.1
Medical (2 epochs)	3,513	439	54.9	34.7
Pooled (all data)	47,450	3,184	316.6	34.7
Normalization, LoRA 16-bit	24,000	1,500	60.0	18.1

dialect sentence x_i and its Standard translation y_i embed close together, with the symmetric InfoNCE loss [22] over a batch of $B = 32$ pairs:

$$\mathcal{L} = -\frac{1}{2B}\sum_{i=1}^{B}\left[\log\frac{e^{s_{ii}/\tau}}{\sum_j e^{s_{ij}/\tau}} + \log\frac{e^{s_{ii}/\tau}}{\sum_j e^{s_{ji}/\tau}}\right], \tag{4.7}$$

where $s_{ij} = \cos(\mathbf{e}_{x_i}, \mathbf{e}_{y_j})$, $\tau = 0.05$, 600 steps and learning rate 2×10^{-5}. Every third batch replays (question, gold passage) pairs so that retrieval ability is not forgotten; the encoder is evaluated exactly like the others.

4.6.7 Routed Experts and the Pooled Control

Three answer experts (education, general, medical) are trained on their domain's QA data with three retrieved passages in the prompt (Table 4.7). As a control, one *pooled* adapter is trained on the union of all expert data with the same total number of optimizer steps as all experts together (3,184 steps, 317 vs. 316 minutes), so "routed experts vs. one adapter" is compared at equal compute. Table 4.8 lists the shared hyper-parameters.

Table 4.8: Shared training and decoding settings.

Setting	Value
Base model	Gemma-4-E4B-it, 4-bit NF4, double quantization, BF16 compute
LoRA	$r = 16$, $\alpha = 32$, dropout 0.05, all attention and MLP projections
Optimizer	AdamW [26], learning rate 2×10^{-4}, 3% warm-up, cosine decay
Batch	4 examples $\times$ 4 accumulation steps = 16
Sequence limit	4,096 tokens (0 examples dropped)
Seed	17 (finalists of S2: 17, 42, 101)
Decoding	greedy; 96 new tokens for normalization (up to 384 for long questions), 64 for education/general, 200 for medical
Evidence	$k = 3$ passages, best first

4.7 Evaluation Protocol

4.7.1 Metrics

Normalization uses chrF++ as the primary metric [17], because character n-grams tolerate the spelling variation typical of dialects:

$$\mathrm{chrF}_\beta = (1+\beta^2)\frac{\mathrm{chrP}\cdot\mathrm{chrR}}{\beta^2\,\mathrm{chrP}+\mathrm{chrR}}, \quad \beta = 2, \tag{4.8}$$

where chrP and chrR are the character n-gram ($n \le 6$) precision and recall; chrF++ adds word unigrams and bigrams. BLEU [8] (computed with sacreBLEU [23]), character error rate and, for comparisons with published work, ROUGE [10] and METEOR [11] are reported as well.

Extractive QA (education, general) uses exact match and token F1:

$$F1 = \frac{2PR}{P+R}, \quad P = \frac{|\hat{a}\cap a|}{|\hat{a}|},\ R = \frac{|\hat{a}\cap a|}{|a|}, \tag{4.9}$$

over the bags of tokens of the predicted answer $\hat{a}$ and the gold answer a (maximum over gold answers). **Medical** answers use chrF++ and ROUGE-L against the doctor's answer, plus the referral rate and the share of answer words found in the evidence. **Retrieval** uses hit@k, MRR@10 and nDCG@10 at document level, and *answer-containment* hit@k (the gold answer string occurs in the top-k passages). **Routers** use accuracy, macro-F1 and ECE.

The end-to-end **QA score** of a question is its token F1 (education, general) or chrF++/100 (medical), and 0 when an answerable question is not answered; behaviour is scored separately (answered share, correct abstention on unanswerable and out-of-scope questions, clarification on vague ones). The **dialect gap** is $g_v = \mathrm{QA}_{\mathrm{Std}} - \mathrm{QA}_v$, and the share of the gap a system closes is

$$\mathrm{closed}_v = \frac{g_v^{\mathrm{NO_NORM}} - g_v^{\mathrm{system}}}{g_v^{\mathrm{NO_NORM}}}. \tag{4.10}$$

4.7.2 Statistics

All tables report 95% percentile-bootstrap confidence intervals over test items [4], [9]. Two systems evaluated on the same items are compared with a paired

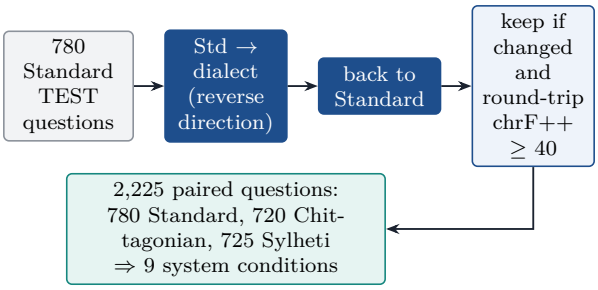


Figure 4.16: Building the paired test set. Because the dialect versions are model-written and "lighter" than real dialect text, a second evaluation uses 481 questions written by native speakers (Section 4.7.4).

approximate-randomization test [12]: the per-item scores of the two systems are randomly swapped 2,000–5,000 times, and p is the share of swaps whose mean difference is at least as large as the observed one. The full system is compared with seven ablations in three varieties plus the oracle, giving 22 p-values, which are corrected with the Holm procedure [2]: with $p_{(1)} \leq \cdots \leq p_{(m)}$, the i-th hypothesis is rejected while $p_{(i)} \leq \alpha/(m - i + 1)$.

4.7.3 Paired End-to-End Test Set

Dialect QA test data with reference answers do not exist. We therefore take 780 test questions (200 each from education, general and medical, plus 60 unanswerable, 60 out-of-scope and 60 under-specified), rewrite each into Chittagonian and Sylheti with the reverse direction of the normalization expert, and keep a dialect version only if it differs from the original and a round trip back to Standard reaches chrF++ ≥ 40 (Fig. 4.16). 92.3% of the Chittagonian and 92.9% of the Sylheti versions were kept, giving 2,225 questions: the *same* questions in three varieties, so any score difference is caused by the variety alone.

4.7.4 Human-Written Dialect Questions

The question sentences in the held-out test split of the native-written parallel corpora (Kothon, Vashantor, ChatgaiyyaAlap, ONUBAD) give 481 real dialect questions (234 Chittagonian, 247 Sylheti), each with its human Standard translation but without a reference answer. The system is therefore evaluated by *consistency*: does it treat the dialect question like its Standard version (same decision, same domain, similar answer, overlapping evidence)?

4.7.5 Ablation Conditions

Table 4.9 lists the nine conditions run on the paired test set. Each removes or replaces one component, so the difference to `FULL` measures that component's contribution.

Table 4.9: End-to-end evaluation conditions.

Condition	What is changed	Answers
FULL	nothing (proposed system)	per variety
NO_NORMALIZATION	dialect text goes straight to routing, retrieval, expert	RQ2
ALWAYS_NORMALIZE	dialect router bypassed; everything normalized	router value
NO_DOMAIN_ROUTING	one merged KB and the pooled adapter	RQ3
POOLED_ADAPTER	routed experts → one pooled adapter	RQ3
BASE_MODEL	no fine-tuning (same prompts)	RQ3
NO_RAG	closed-book answering	RQ3
NO_CONTROLLER	always answer	RQ3
ORACLE_STANDARD	gold Standard question and domain	upper bound

4.7.6 Comparability with Published Work

To compare with published dialect-translation models, the normalization expert is also scored on the *official* Vashantor test sentences [51] with the papers' own metrics (BLEU, BLEU-1, chrF, METEOR, WER, CER) and 95% bootstrap intervals; an audit confirms that these sentences are the official test split and never occur in training. The QA experts are compared with the gold-passage results of BanglaRQA and NCTB-QA on the subset of questions whose gold passage was retrieved.

4.7.7 Answer-Time Improvements (System v2)

After the main results were frozen, a set of improvements that change only *how* the trained models are used was defined (Table 4.10, Fig. 4.17). Each was switched on only if it improved the DEV split by a fixed margin (0.005 for retrieval and QA scores). The paired TEST set was then answered once more: a run with every change switched off, on 300 sampled test questions, checks that the v2 code reproduces the reported answers (validity check), and the adopted changes are added one at a time so that each step has its own paired test. The main results stay as reported; v2 is an extension.

The last column of Table 4.10 records the DEV decisions. Three changes were adopted: re-ranking of 50 candidates for education and general questions, with near-duplicate removal for education only (I4); searching both the education and the general knowledge base when the domain router's confidence is below $\tau = 0.9$, which happened for about 46% of DEV questions (I5); and the routing-independent medical note (I3). The other five were tested and not adopted, because they did not reach the DEV margin. The glossary helped Chittagonian but hurt Sylheti. Dual-query retrieval changed answer containment by less than 0.005. A forced second decoding did not raise the DEV QA score for any threshold. Repetition trimming lowered it slightly. The trigger chosen on DEV was the S5 rule itself.

Table 4.10: Answer-time improvements (no retraining), each chosen on DEV.

	Change	DEV criterion	DEV decision
I1	also normalize "Standard" predictions with confidence $< t$	Youden's J [1] on dialect-ID DEV	not adopted: the S5 rule had the highest J (0.958)
I3	medical note and dose refusal whenever a medical signal fires, independent of routing	F2 on domain DEV	**adopted**: domain-router medical probability ≥ 0.2
I4	50 candidates, near-duplicate removal [44], cross-encoder re-ranking [27], [60]	answer-containment hit@3	**adopted**: education re-rank + de-duplication (0.730 $\rightarrow$ 0.812); general re-rank (0.727 $\rightarrow$ 0.749)
I5	search education and general KBs when the domain router's confidence $< \tau$ [85]	answer-containment hit@3	**adopted**: $\tau = 0.9$ (0.697 $\rightarrow$ 0.767)
I6	when the expert says "not found", decode again with that sentence blocked; keep it if mean log-prob. $\geq \theta$	DEV QA, abstention -5 pts max	not adopted: DEV QA unchanged (0.540) at every θ
I7	cut an answer where it starts repeating itself [47]	DEV QA not lower	not adopted: 0.540 $\rightarrow$ 0.539
I8	dialect glossary (word entries, TRAIN split) shown to the normalization expert as example turns	chrF++ $\geq$ +0.5, no variety lower	not adopted: Chittagonian +0.55, Sylheti -0.73
I9	retrieve with original *and* normalized question, fused by RRF [73]	hit@3 on dialectized DEV	not adopted: mean gain < 0.005

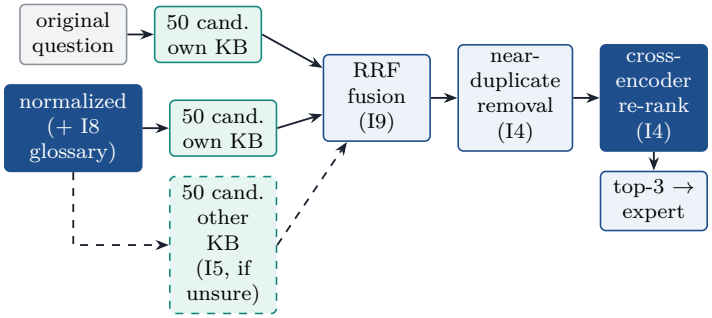


Figure 4.17: Retrieval in system v2 as designed. After the DEV decisions (Table 4.10), the adopted path is: normalized question $\rightarrow$ 50 candidates from the domain's KB, plus 50 from the other KB when the domain router's confidence is below 0.9 (I5) $\rightarrow$ near-duplicate removal (education) $\rightarrow$ cross-encoder re-ranking $\rightarrow$ top 3. The glossary (I8) and RRF fusion (I9) were tested but not adopted.

4.8 Implementation Environment

The implementation is a Python 3.12 library of 20 modules (data, metrics, retrieval, routers, prompts, training and inference, pipeline, analysis, reporting) embedded in the stage notebooks, so each notebook runs on a fresh machine. The GPU stack is PyTorch 2.8 (CUDA 12.8), Hugging Face Transformers 4.57 (and 5.x in a second environment for Gemma-4), PEFT 0.21, bitsandbytes 0.50 and Accelerate 1.15; the CPU stack adds scikit-learn 1.6 and sacreBLEU 2.5. GPU jobs run detached with atomic checkpoints (adapter, optimizer, scheduler, data order and random states), so a stopped machine resumes where it stopped. Fig. 4.18 shows the GPU time per stage.

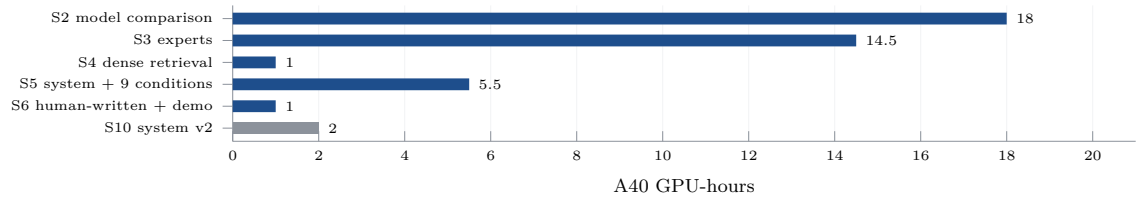


Figure 4.18: GPU time per stage: about 40 A40 GPU-hours (≈US$20 at US$0.49/h) up to S6; system v2 added about 2 more hours.

4.9 Summary

The methodology turns a Standard-only QA problem into a dialect-aware one with small, inspectable parts: a CPU dialect router decides *whether* to rewrite, one LoRA expert rewrites, a CPU domain router and controller decide *where* and *whether* to answer, each domain retrieves its own evidence, and one LoRA expert per domain answers from it on a single shared 4-bit model, with deterministic rules for emergencies and medicines. Every decision is made on validation data, every result is reported per language variety, and the evaluation isolates the effect of each component: the same questions in three varieties, nine conditions with Holm-corrected paired tests, native-speaker questions, comparisons with published work on the same test sentences, and a DEV-selected set of answer-time improvements. Chapter 5 reports the results.

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